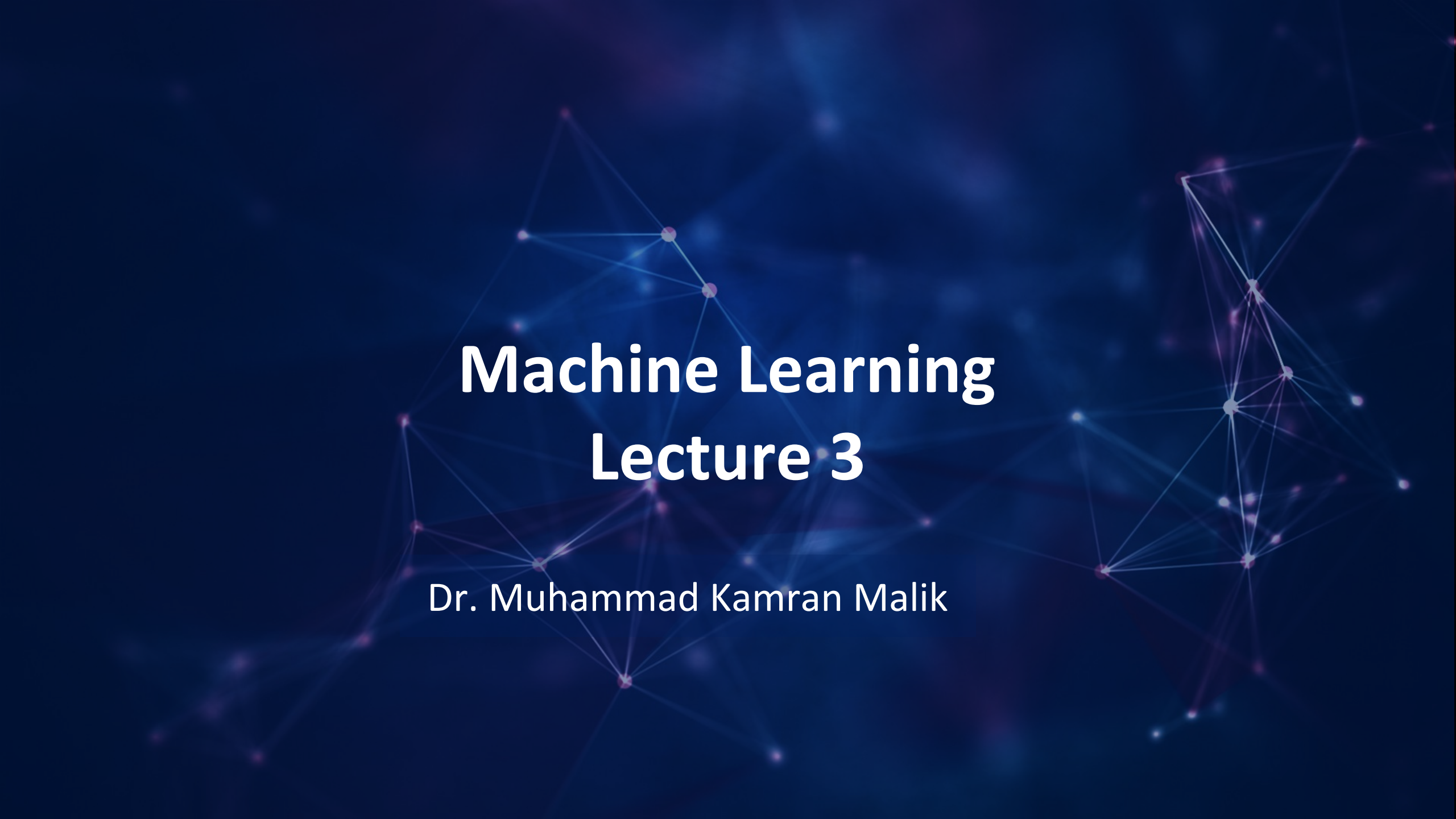




Machine Learning Lecture 3

Dr. Muhammad Kamran Malik

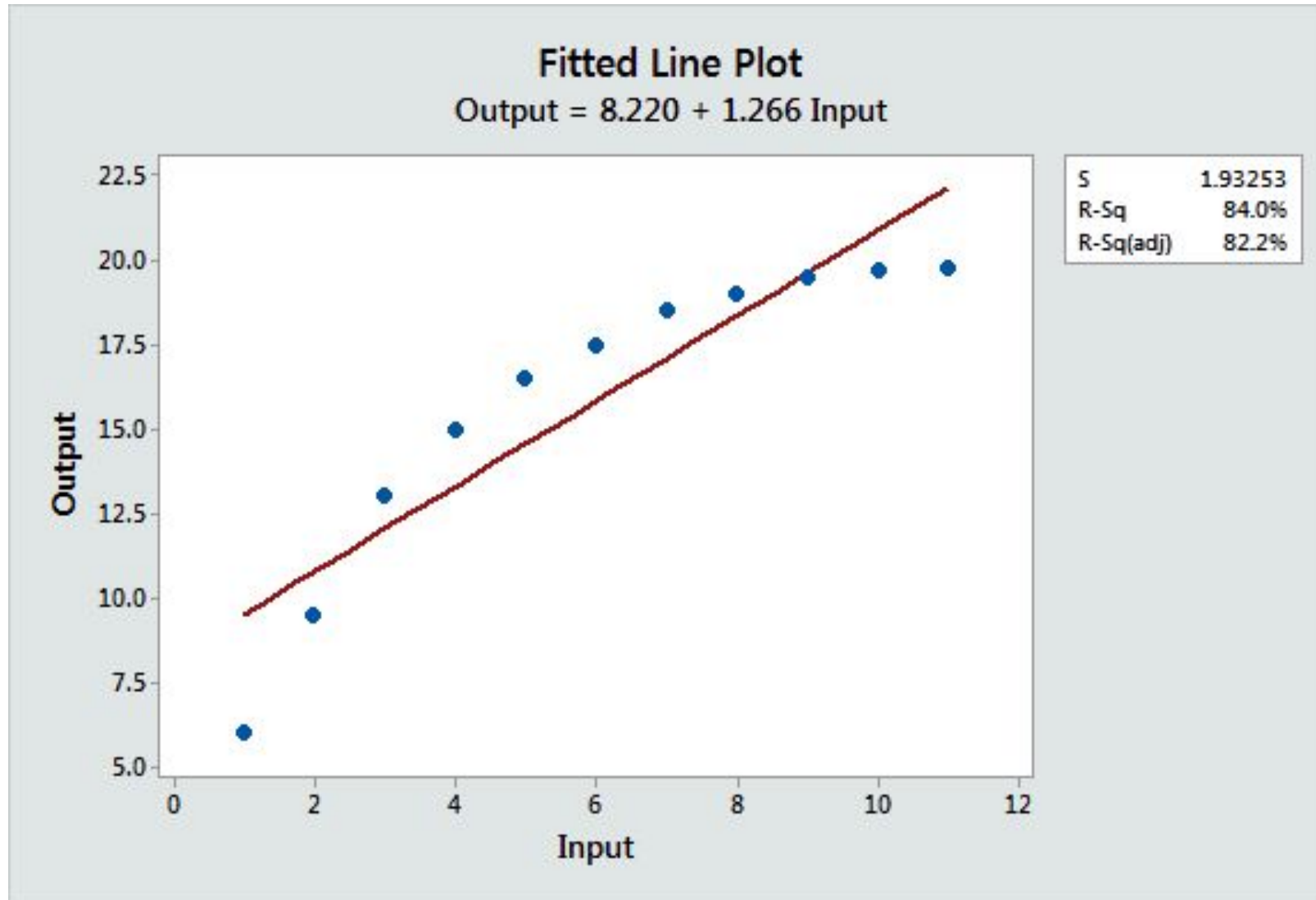


Linear Regression Model



Linear Regression with One Variable

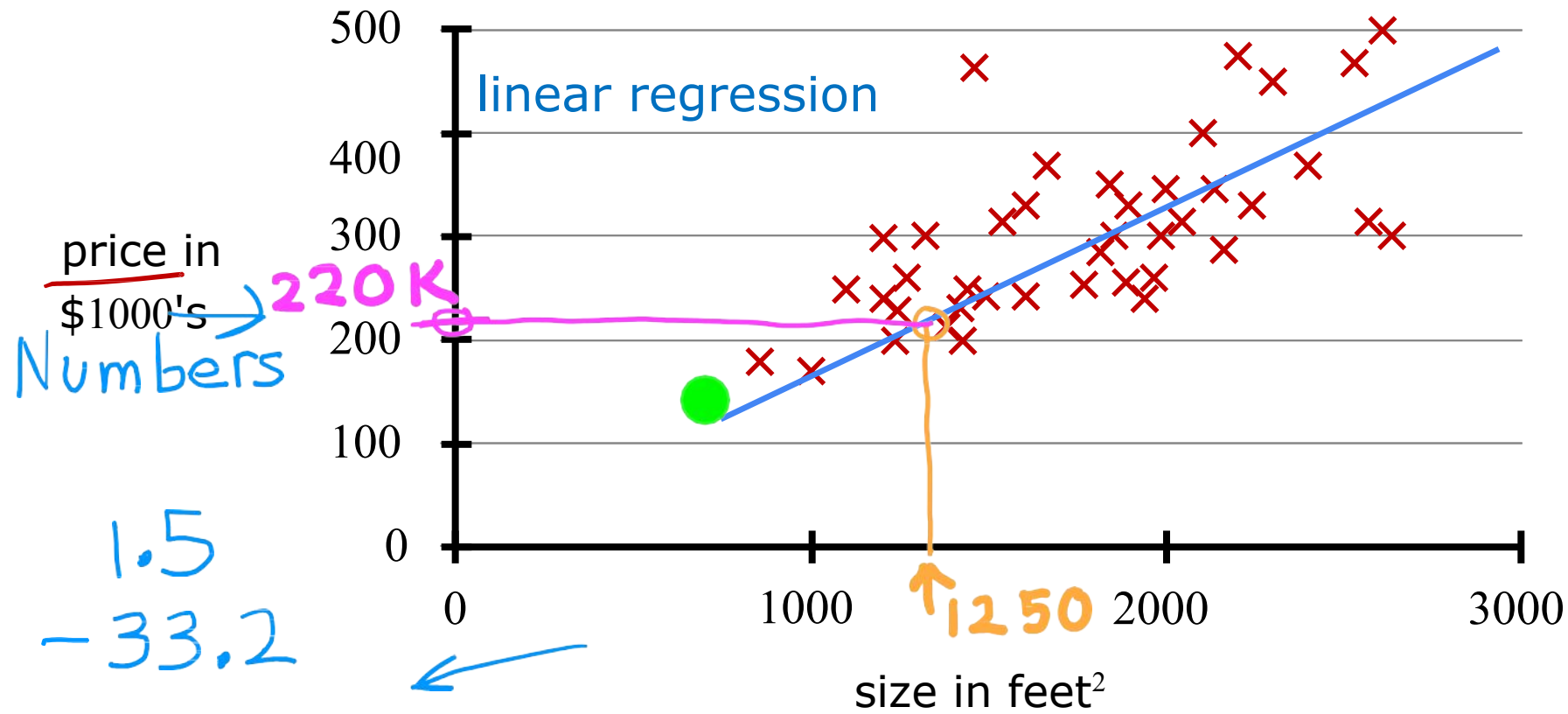
Linear Regression with One Variable



Linear Regression with One Variable

- *A Supervised learning algorithm*
- Taking input variables and trying to fit the output onto a *continuous* values. (*Regression problems*)
- Linear regression with one variable is also known as “Univariate linear regression”.
- Univariate linear regression is used when you want to predict a **single output** value y from a **single input** value x .
- The Hypothesis Function
 - $y' = f_{\vartheta_0, \vartheta_1}(x) = h_{\vartheta}(x) = \vartheta_0 + \vartheta_1 x$
 - $y' = f_{w_0, w_1}(x) = h_{\vartheta}(x) = w_0 + w_1 x$
 - $y' = f_{w, b}(x) = h_{\vartheta}(x) = b + wx$

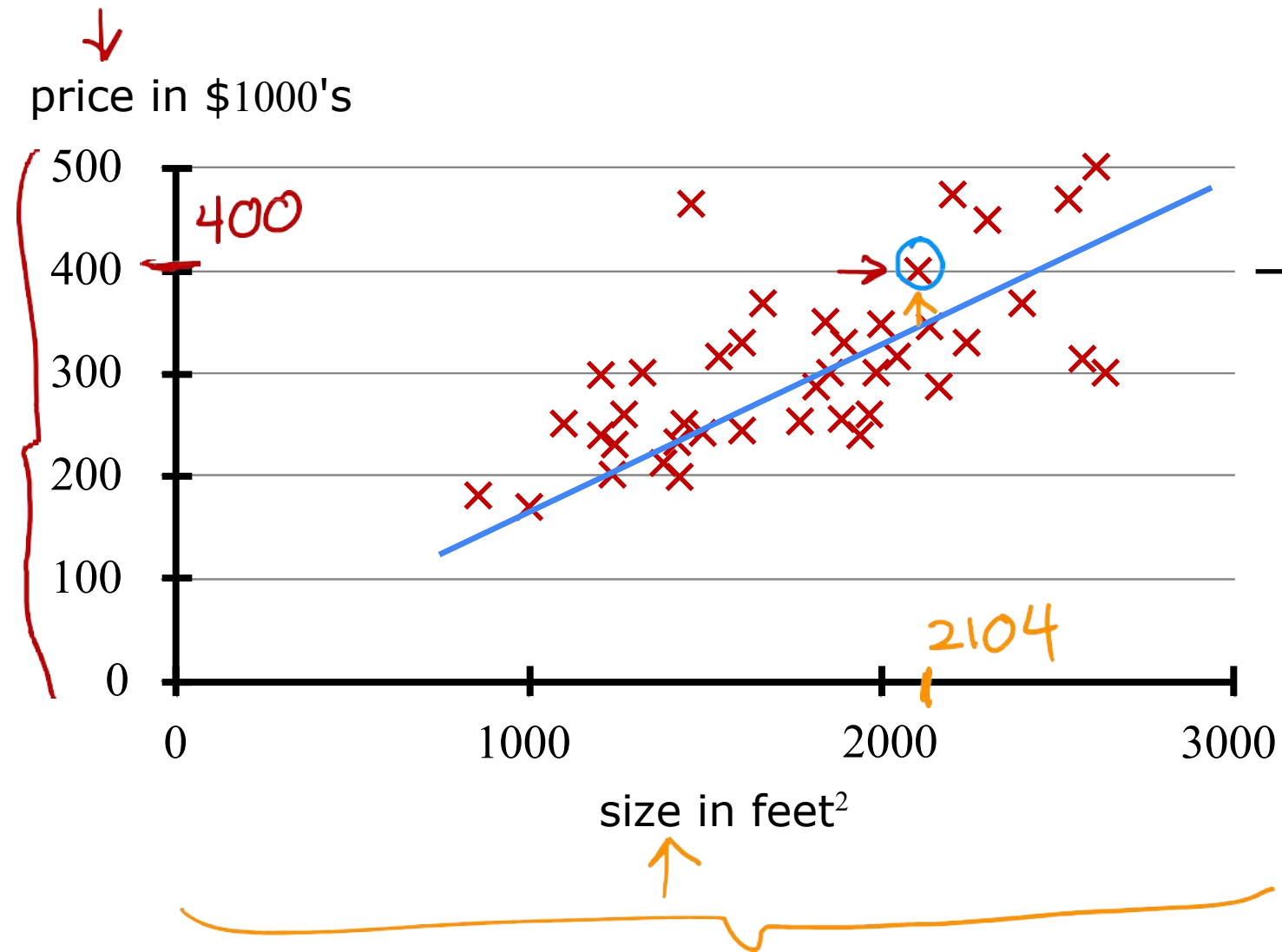
House sizes and prices



Regression model
Predicts numbers
Infinitely many possible outputs

Supervised learning model
Data has "right answers"

House sizes and prices



Data table

size in feet_2	price in \$1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Terminology

Training Data used to train the model

set: x y
→ size in feet² → price in \$1000's

(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$x^{(1)} = 2104$$

$$y^{(1)} = 400$$

$$(x^{(1)}, y^{(1)}) = (2104, 400)$$

$$x^{(2)} = 1416$$

$$x^{(2)} \neq x^2 \text{ not exponent}$$

Notation:

x = "input" variable
feature

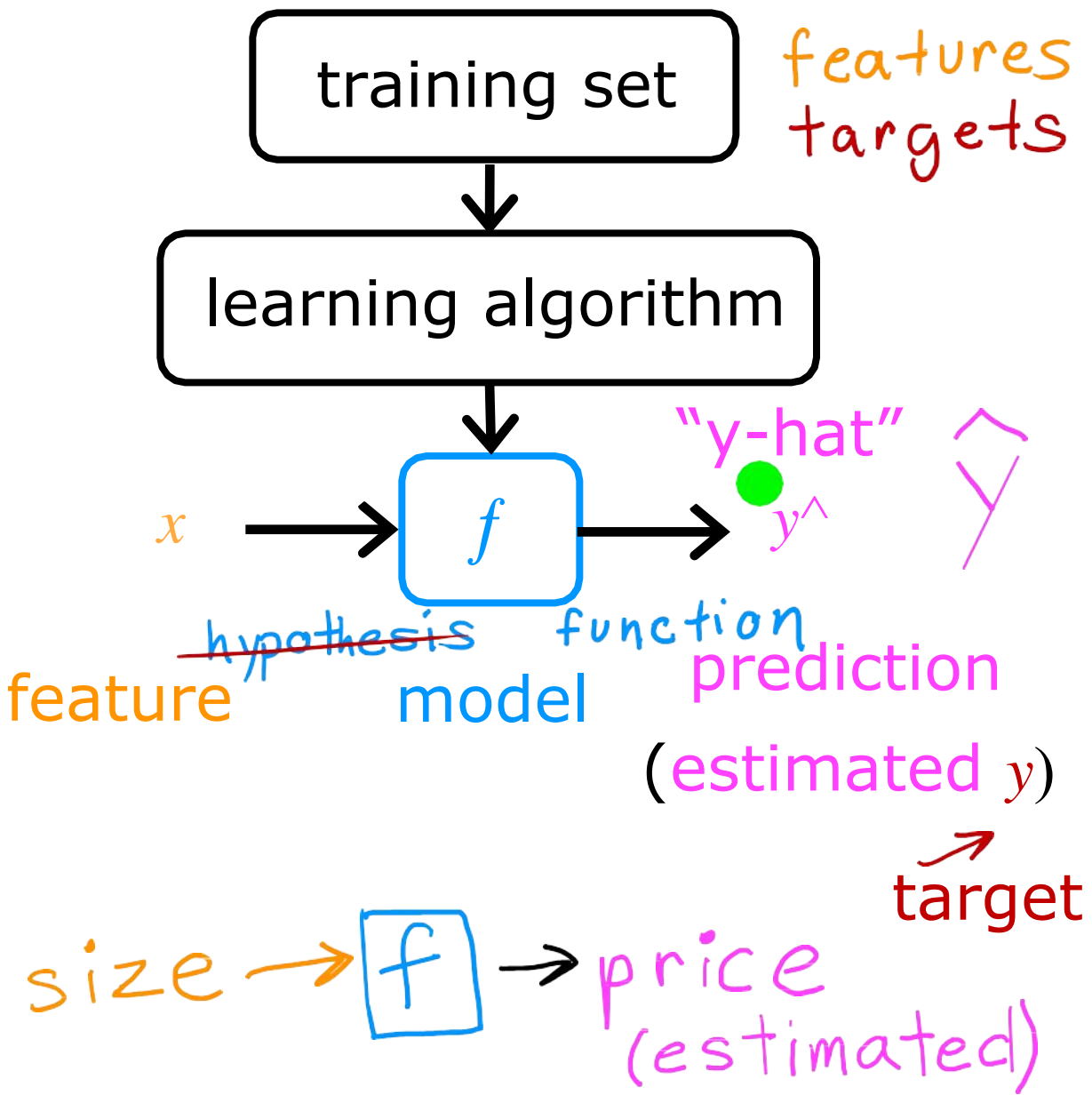
y = "output" variable
"target" variable

m = number of training examples

(x, y) = single training example

$$(x^{(i)}, y^{(i)})$$

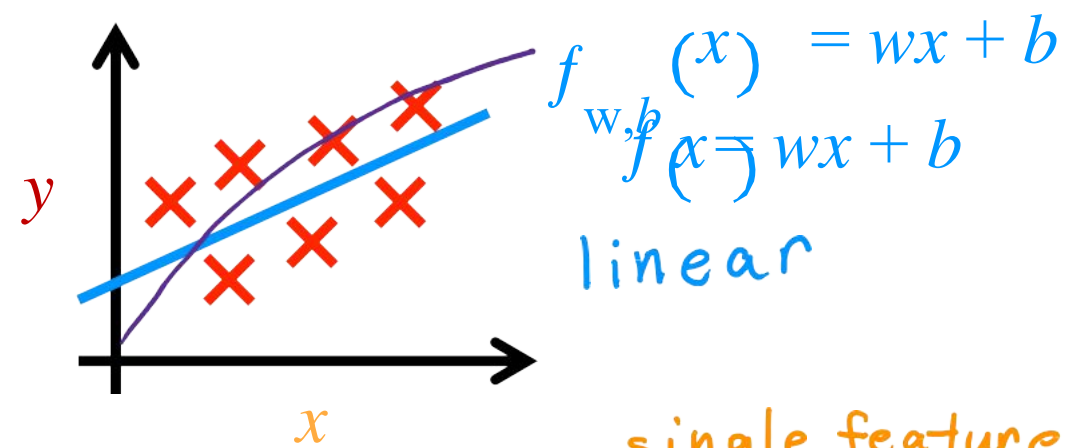
$(x^{(i)}, y^{(i)})$ = i^{th} training example
index (1st, 2nd, 3rd ...)



How to represent f ?

$$f_{w,b}(x) = wx + b$$

$f(x)$



single feature x
with one
size
Linear regression
variable.
Univariate linear regression.
one variable



Q & A



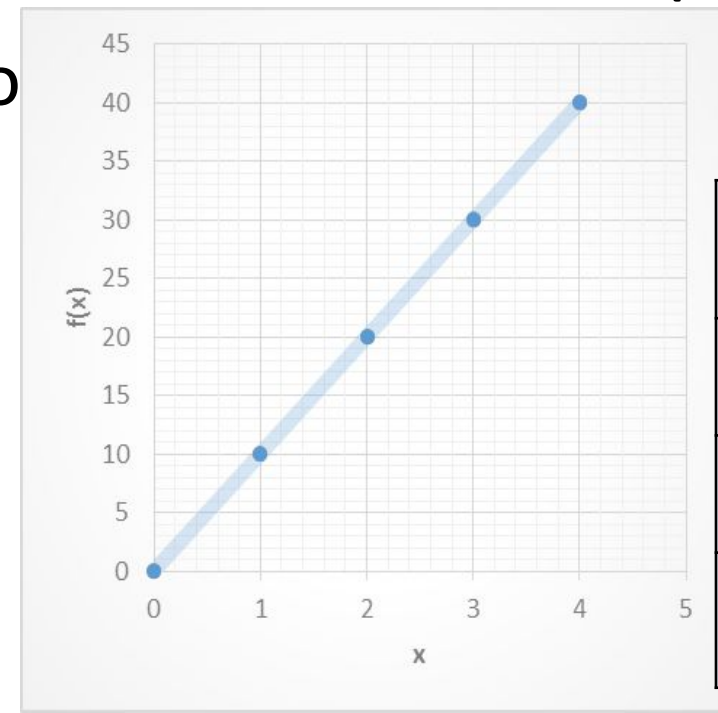
Cost Function

Linear Regression with One Variable

- Note that this is like the equation of a straight line.
- We give to $f_{w,b}(x)$ values for w and b to get our estimated output y' .

- In other words, we are trying to create a function called $f_{w,b}(x)$ that is a (the x's) to o (the y's).

ht	wt	$y1'(5,5)$	$y2'(2,9)$
0	0	5	2
1	10	10	11
2	20	15	20
3	30	20	29
4	40	25	38



ht	wt
5	50
6	60
7	70

Linear Regression with One Variable (Cost Function)

Model $y' = f_{w,b}(x) = wx + b$

Parameters w, b

Cost Function
$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Objective
$$\underset{w,b}{\text{minimize}} J(w, b)$$

Linear Regression with One Variable (Cost Function)

- We can measure the performance of our hypothesis function by using a **cost function**. This takes an average (actually a fancier version of an average) of all the results of the hypothesis with inputs from x 's compared to the actual output y 's.

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m (y'^{(i)} - y^{(i)})^2$$

- This function is called "Mean squared error".
- The mean is halved $1/2m$ as a convenience for the computation of the gradient descent, as the derivative term of the square function will cancel out the $1/2$ term.

Linear Regression with One Variable (Cost Function)

- Now we are able to concretely measure the accuracy of our predictor function against the correct results we have so that we can predict new results we don't have.
- If we try to think of it in visual terms, our training data set is scattered on the x-y plane.
- We are trying to make straight line defined by $f_{w,b}(x)$ which passes through this scattered set of data.
- Our objective is to get the best possible line.

Linear Regression with One Variable (Cost Function)

- The best possible line will be such so that the average squared vertical distances of the scattered points from the line will be the least.
- In the best case, the line should pass through all the points of our training data set.
- In such a case the value of $J(w,b)$ will be 0.

Training set

<i>features</i> size in feet ² (x)	<i>targets</i> price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

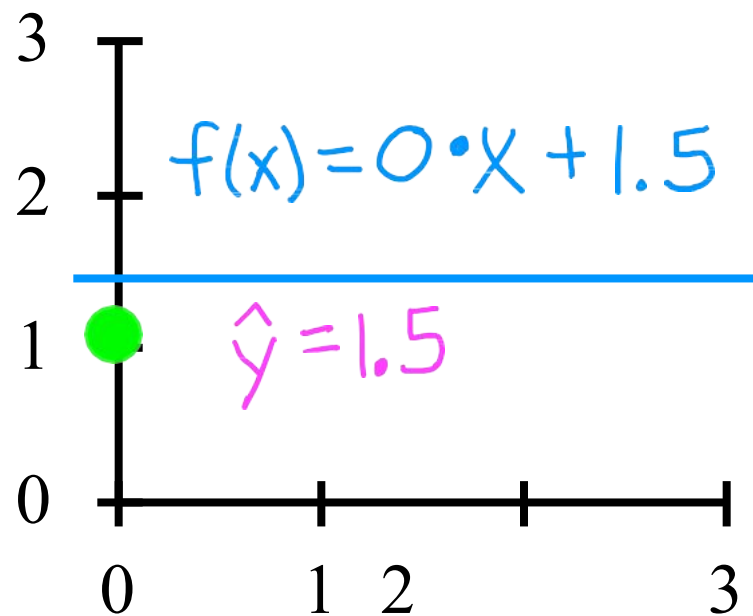
Model: $f_{w,b}(x) = wx + b$

w, b : parameters
coefficients
weights

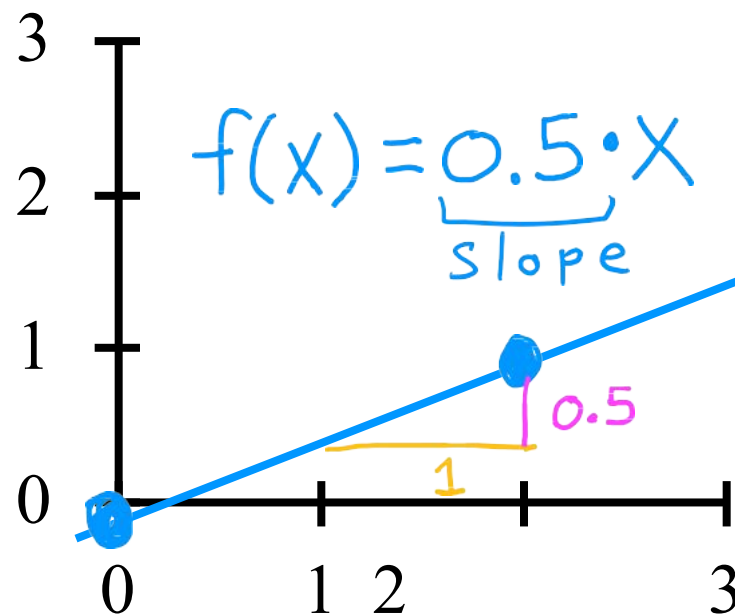
What do w, b
do?

$$f_{w,b}(x) = wx + b$$

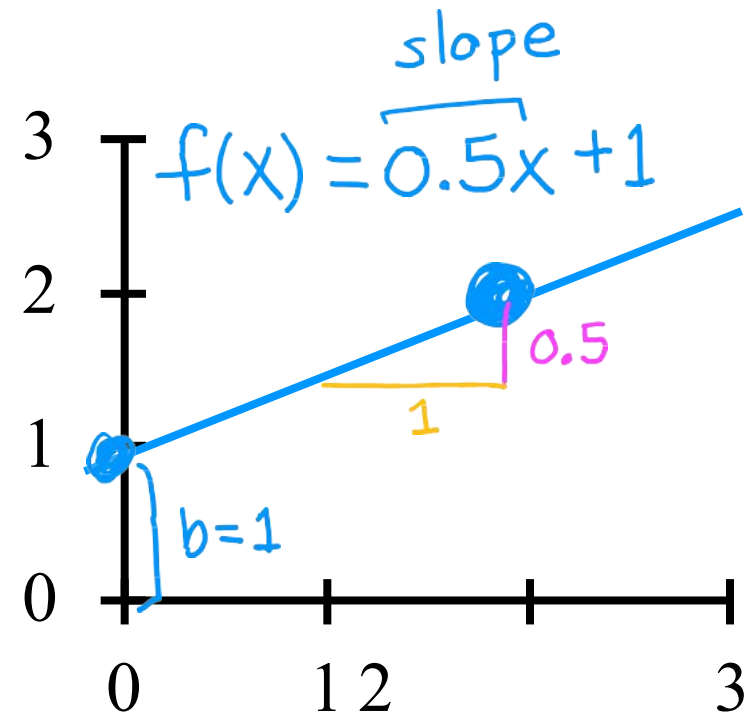
$f(x)$



→ $w = 0$
 → $b = 1.5$
 ↪ y-intercept

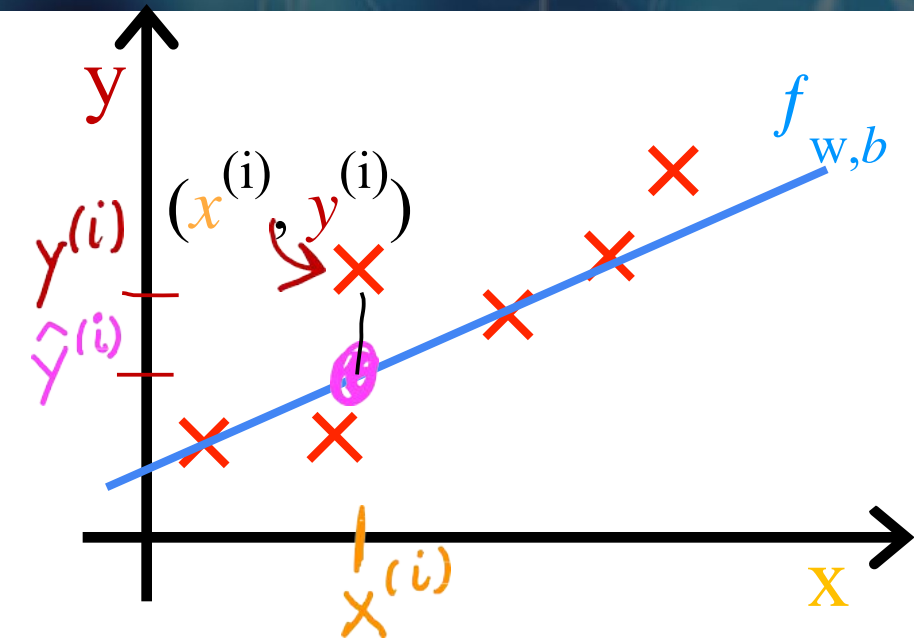


→ $w = 0.5$
 → $b = 0$



→ $w = 0.5$
 → $b = 1$

Cost function: Squared error cost function



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)}) \quad \leftarrow$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m \left(\underset{\substack{\text{error}}}{\hat{y}^{(i)}} - y^{(i)} \right)^2$$

m = number of training examples

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m \left(f_{w,b}(x^{(i)}) - y^{(i)} \right)^2$$

$\uparrow$ intuition (next!)

Find w, b :

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.



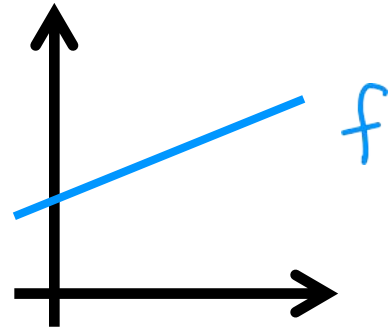
Cost Function Intuition

model:

$$\underline{f_{w,b}}(x) = wx + b$$

parameters:

$$\underline{w, b}$$



cost function:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

$$\underset{w, b}{\text{minimize}} J(w, b)$$

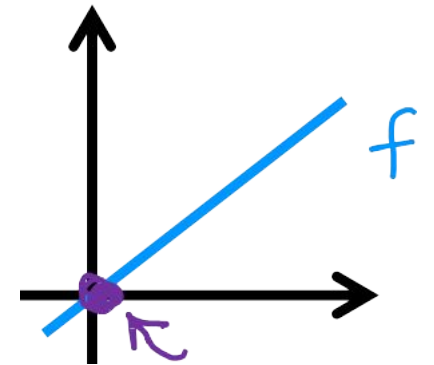
simplified

$$f_w(x) = wx$$

Diagram showing the simplified model equation. A blue arrow points from the variable w below to the coefficient w in the equation. An orange arrow points from the variable x below to the variable x in the equation.

$$b = \emptyset$$

w



$$\underline{J(w)} = \frac{1}{2m} \sum_{i=1}^m (\underline{f_w}(x^{(i)}) - y^{(i)})^2$$

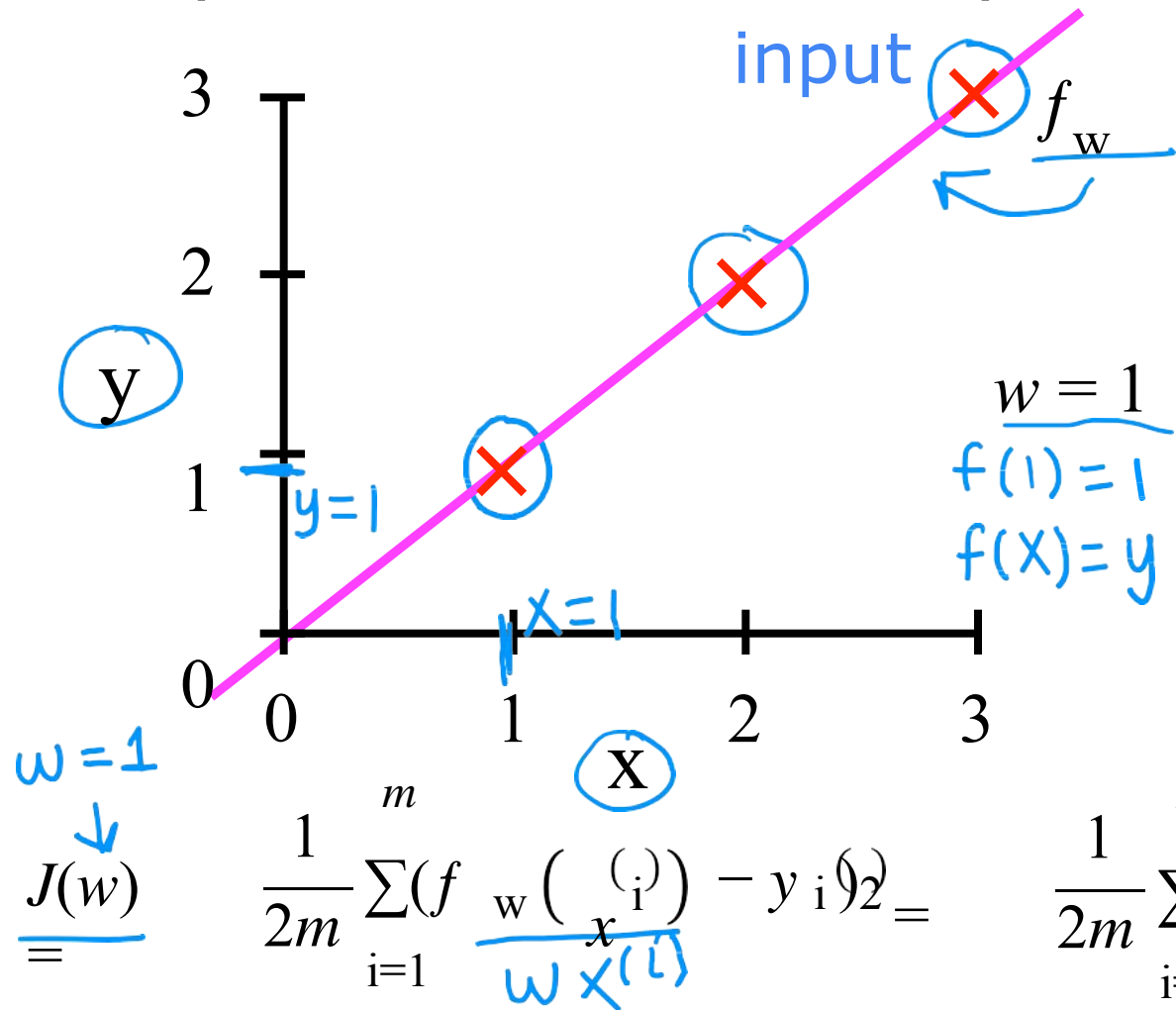
$$\underset{\underline{w}}{\text{minimize}} \underline{J(w)}$$

$$wx^{(i)}$$

A blue arrow points from the expression $w x^{(i)}$ below to the term $f_w(x^{(i)})$ in the equation above.

$f_w(x)$

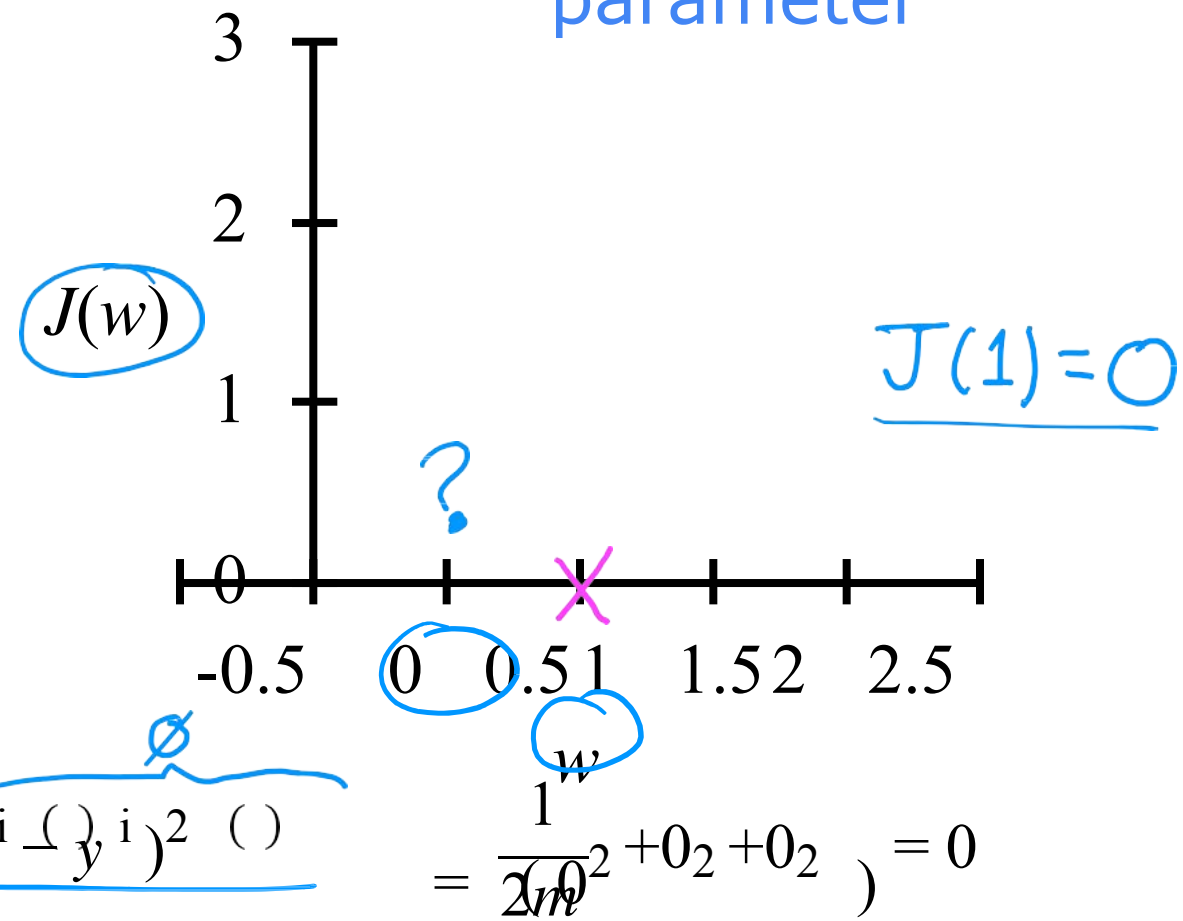
(for fixed w , function of x)



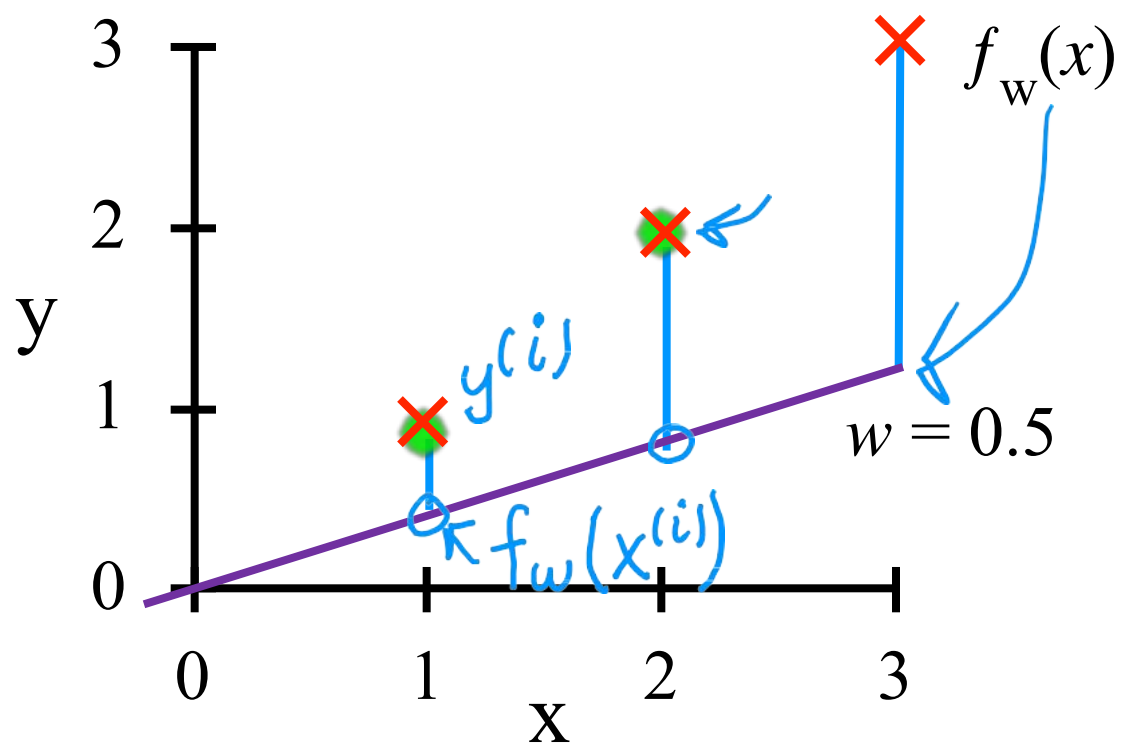
$J(w)$

(function of w)

parameter

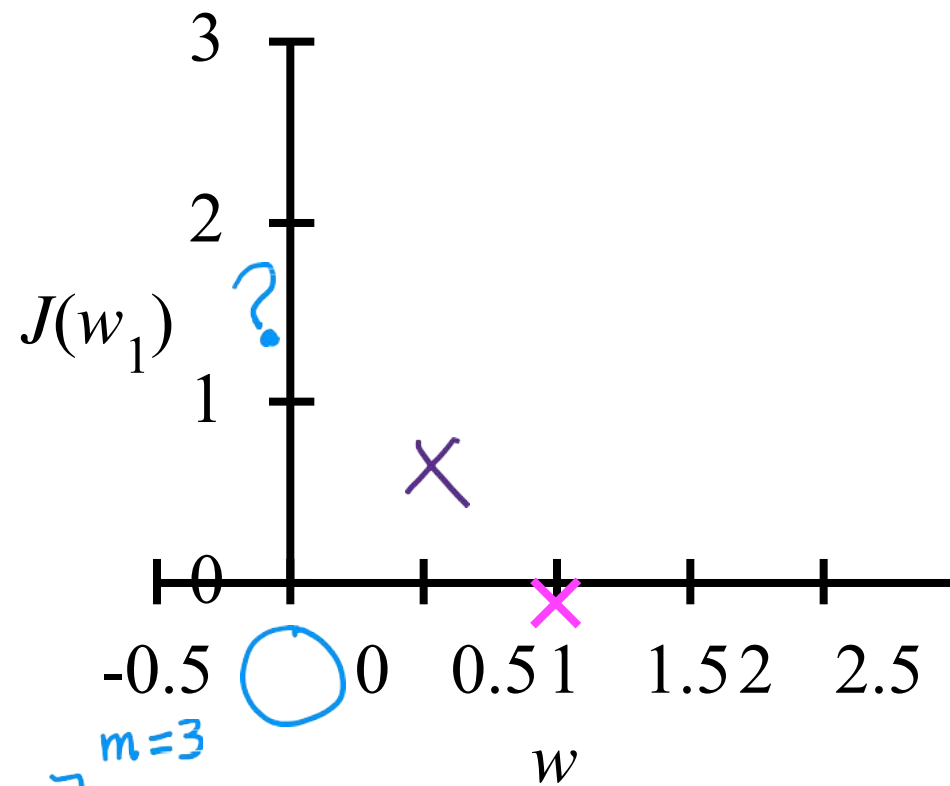


$f_w(x)$
(function of x)

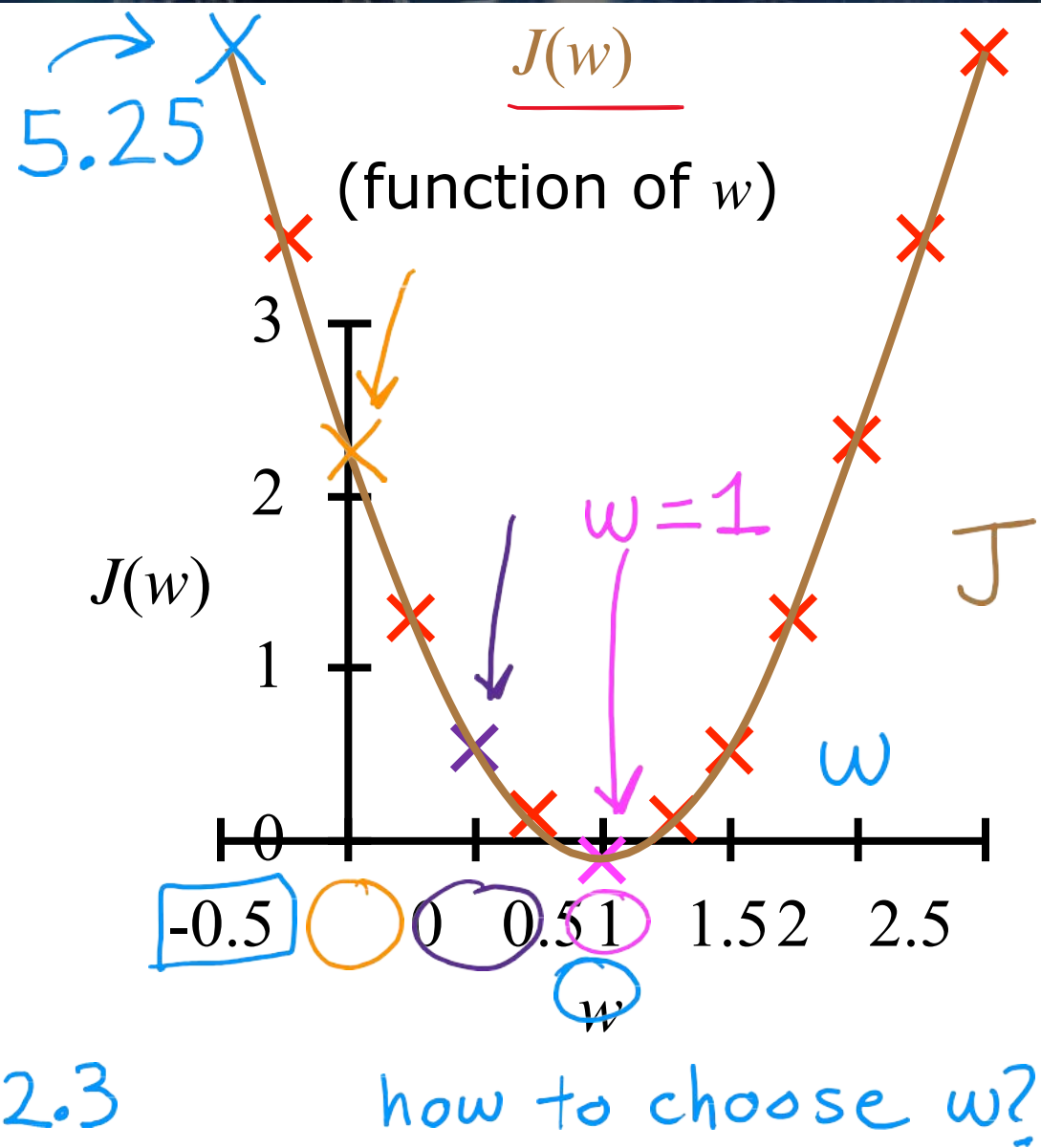
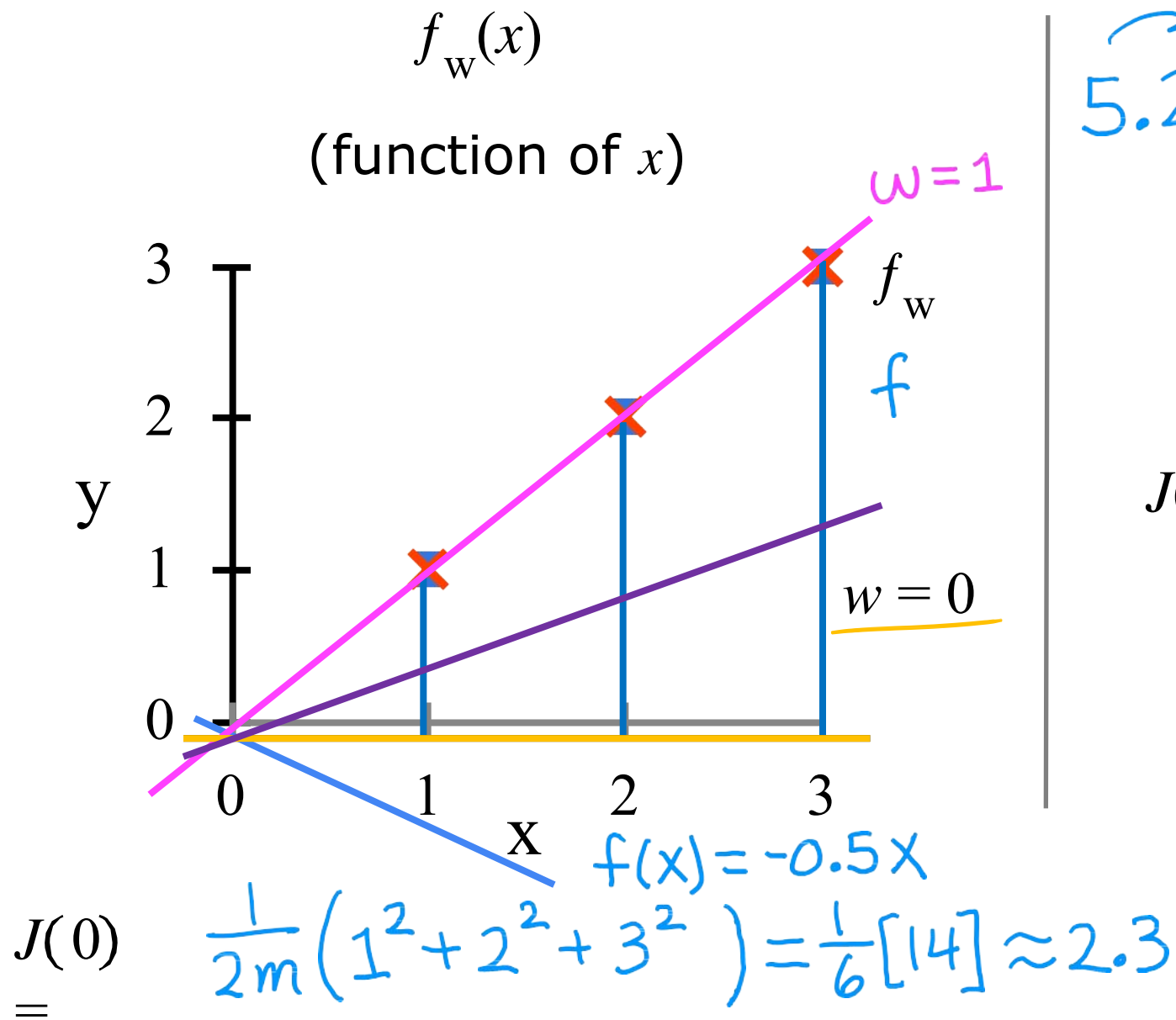


$J(w)$

(function of w)



$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] = \frac{1}{2 \times 3} [3.5] = \frac{3.5}{6} \approx 0.58$$

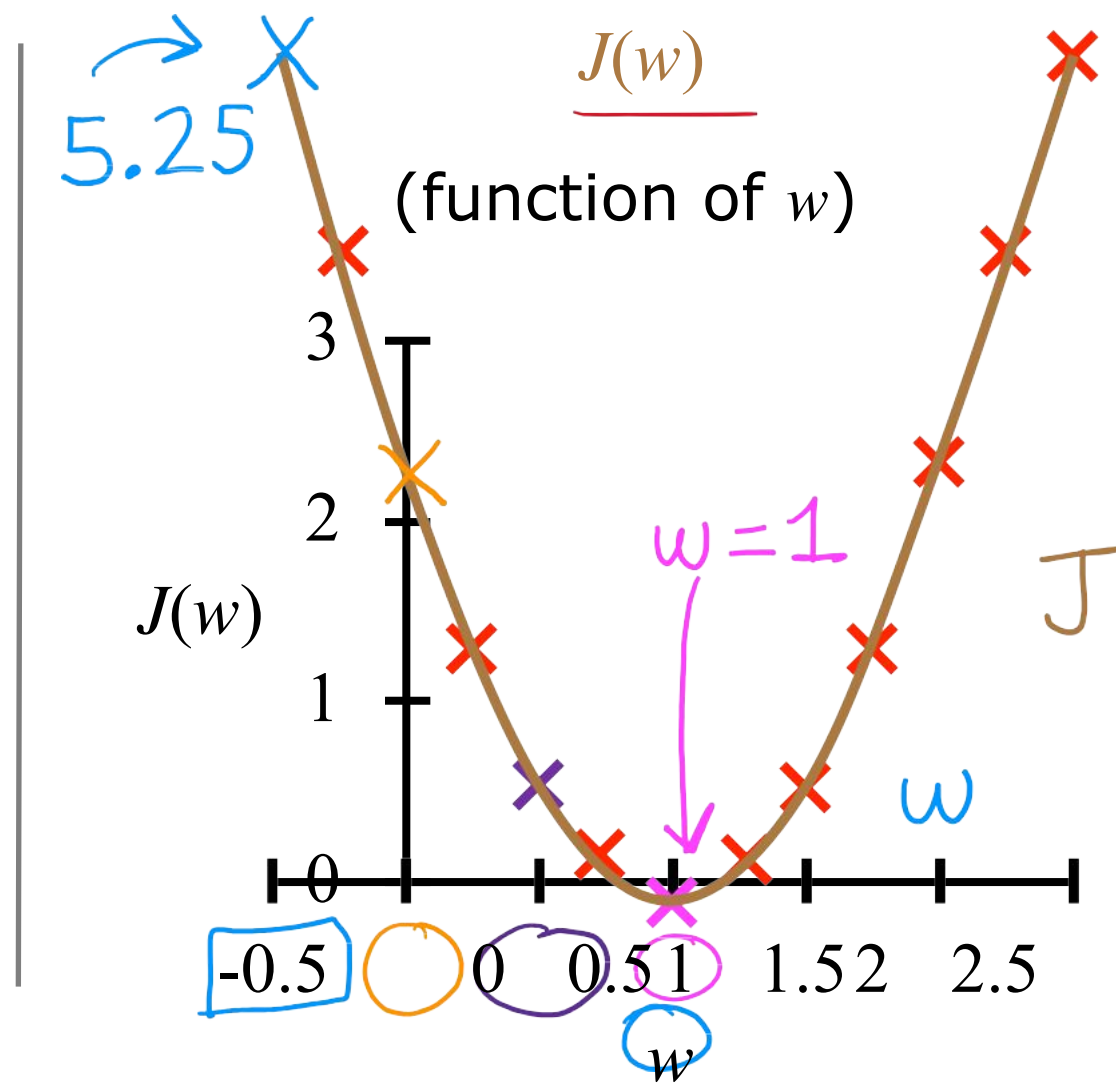


goal of linear regression:

$$\underset{w}{\text{minimize}} J(w)$$

general case:

$$\underset{w, b}{\text{minimize}} J(w, b)$$



choose w to minimize $J(w)$



Visualizing the Cost Function

Model

$$f_{w,b}(x) = wx + b$$

Parameters

w, b

~~before: $b=0$~~

Cost Function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

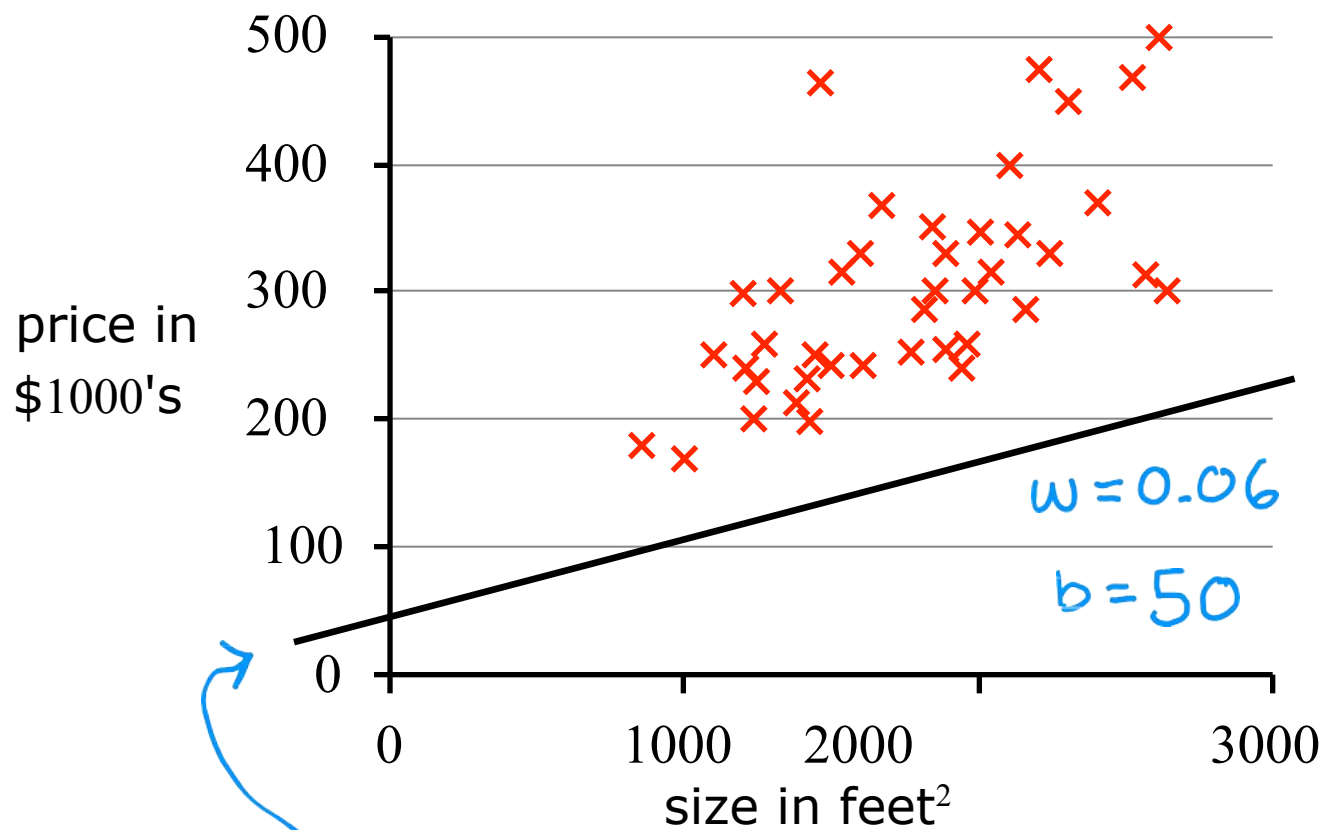
Objective

minimize $J(w, b)$

w, b

$f_{w,b}$

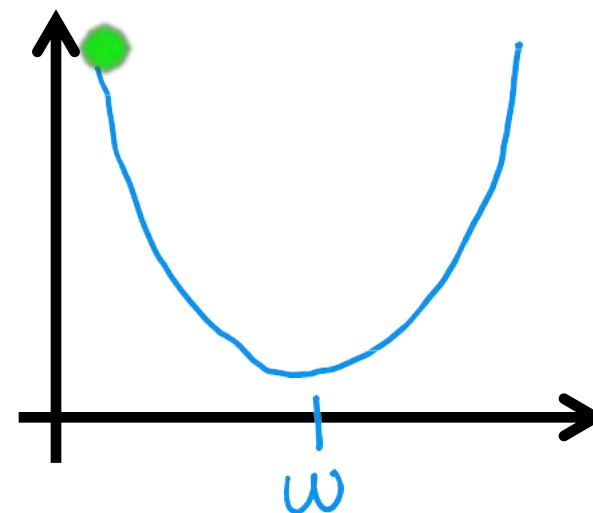
(function of x)



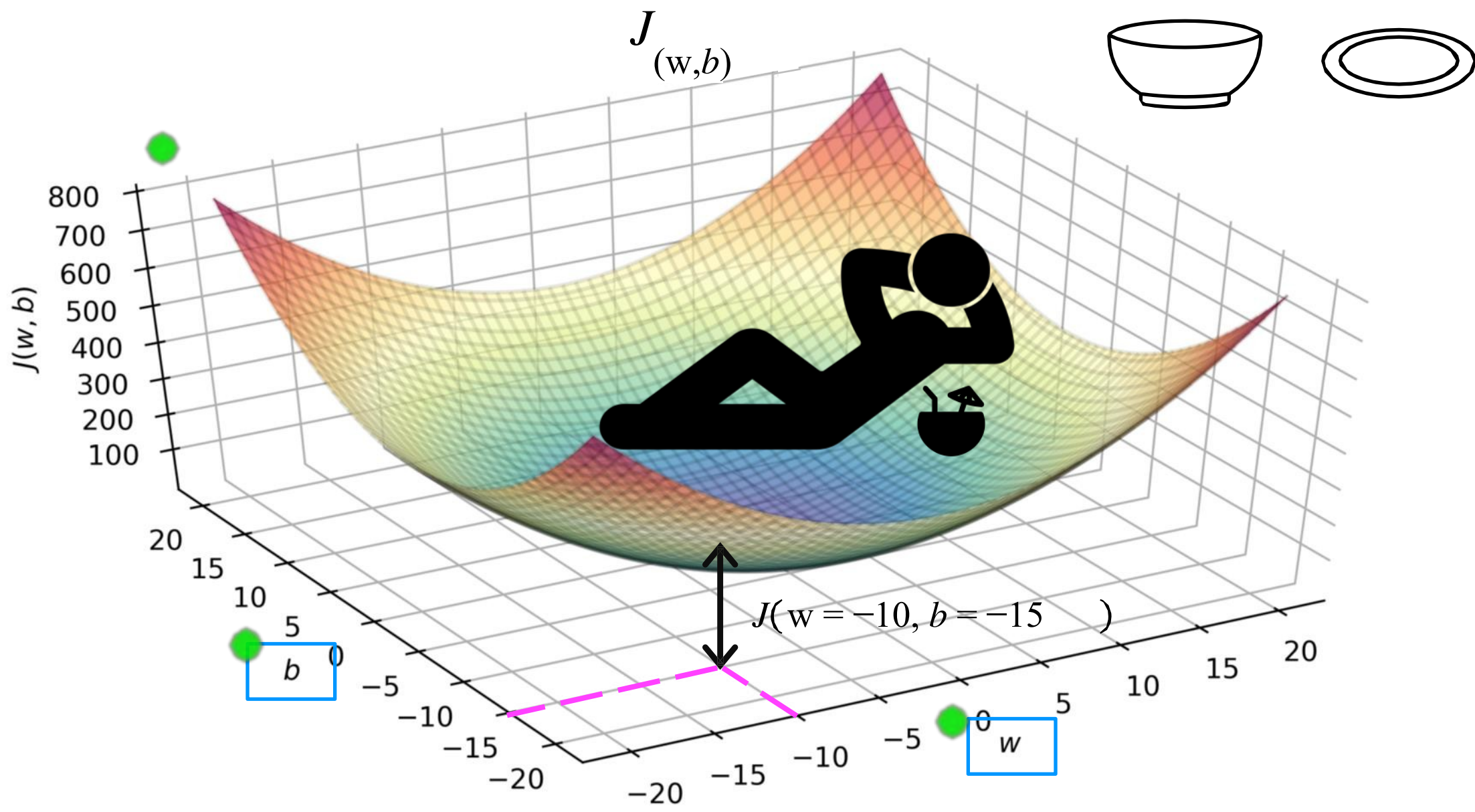
$f_{w,b}(x) = 0.06x + 50$

??

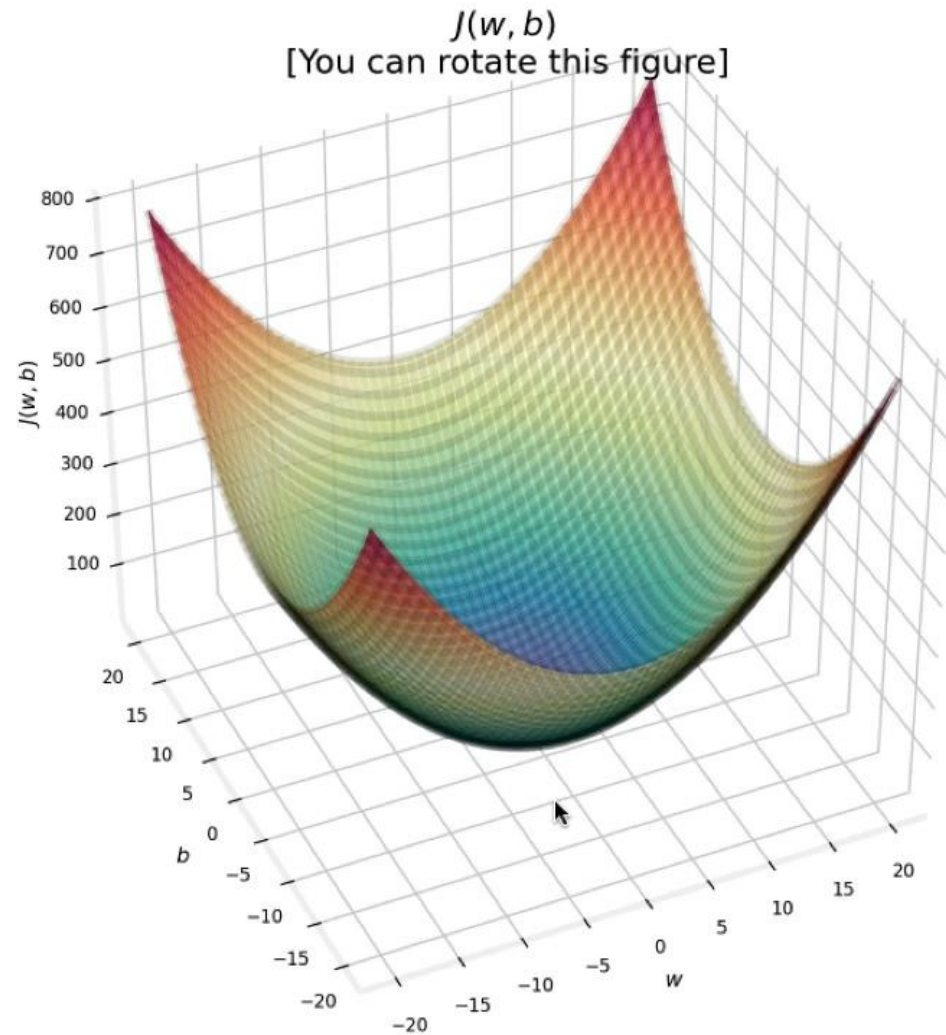
(function of w, b)



w, b



3D surface plot



Alternative
contour plot

Mount Fuji

Legend

- Contour
- Mount Fuji

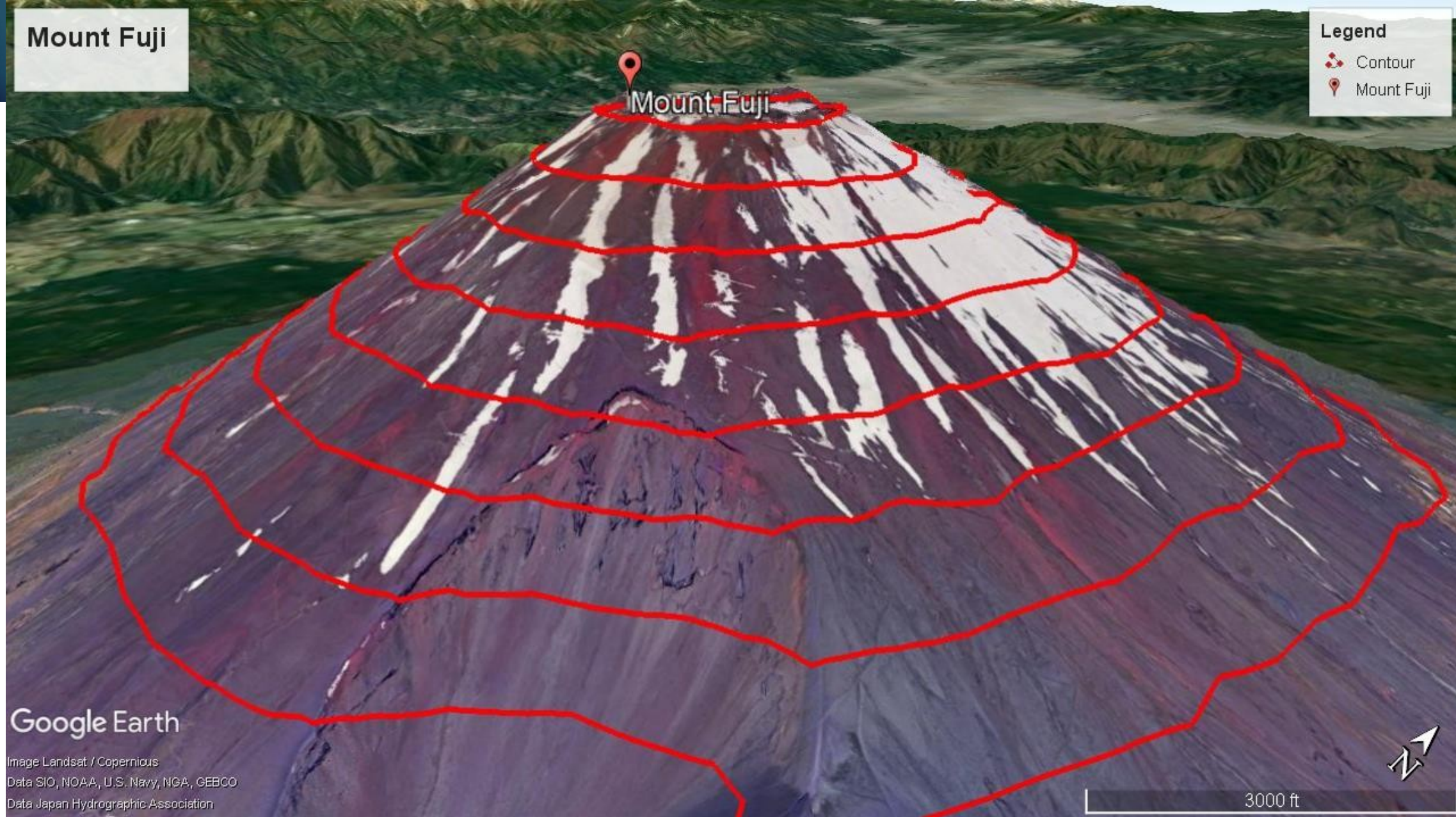
Google Earth

Image Landsat / Copernicus

Data SIO, NOAA, U.S. Navy, NGA, GEBCO

Data Japan Hydrographic Association

3000 ft



Mount Fuji

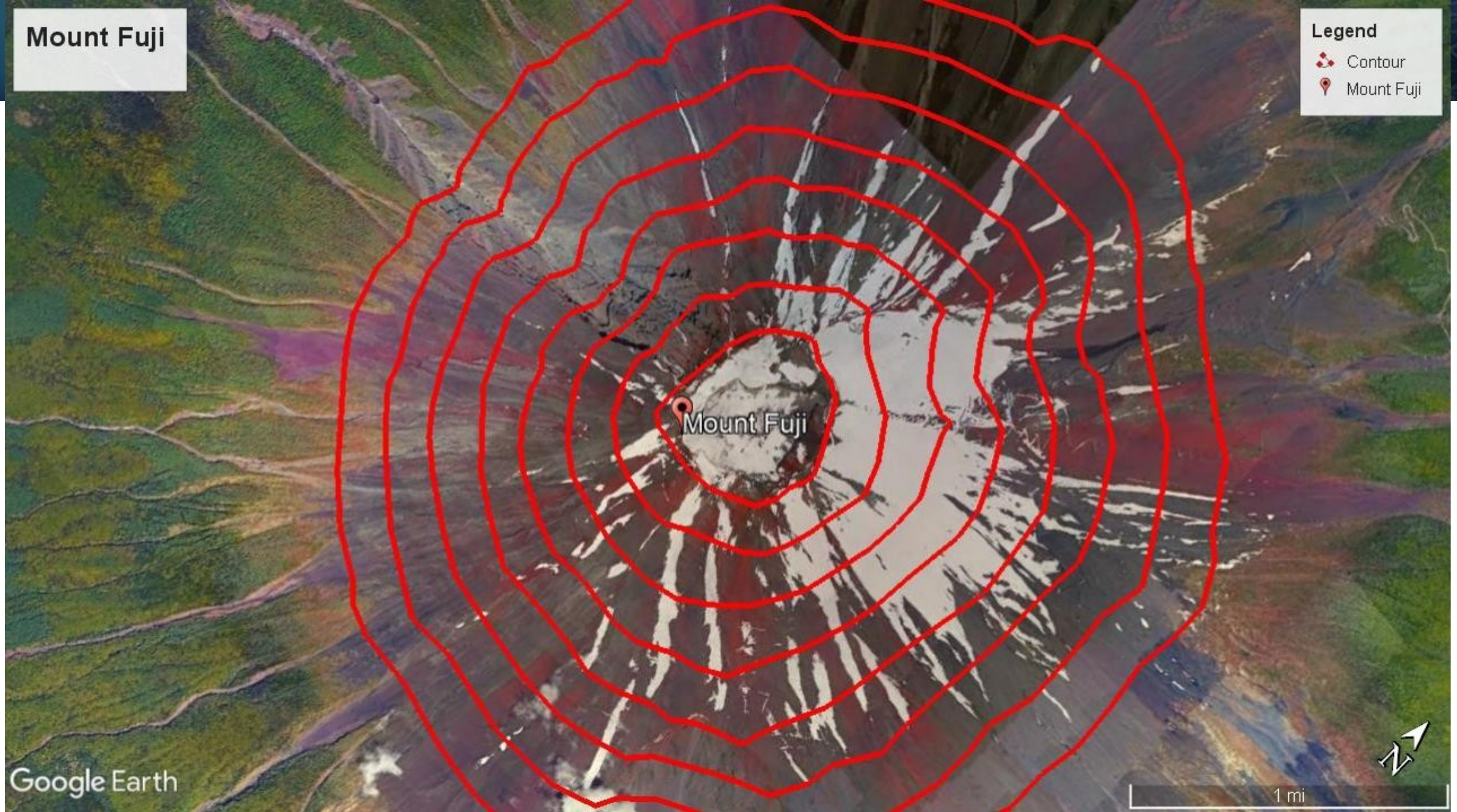
Legend

- Contour
- Mount Fuji

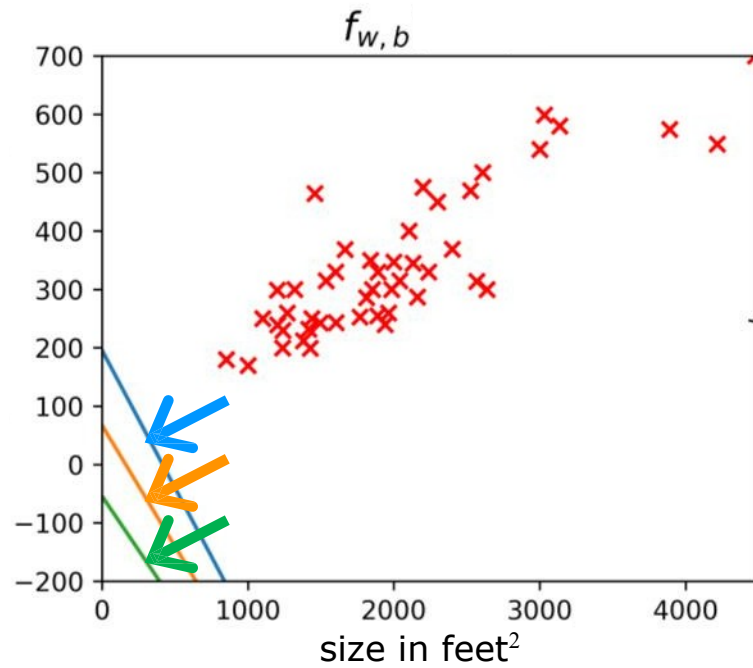
Google Earth

Mount Fuji

1 mi



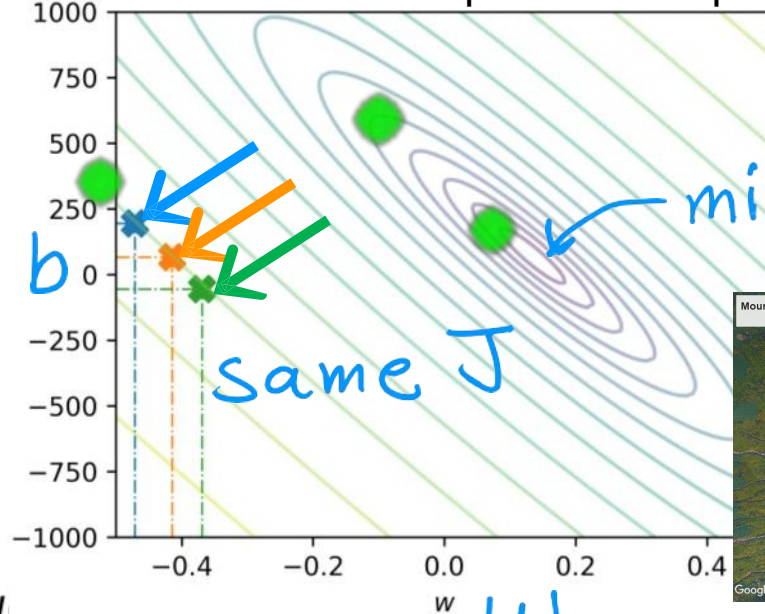
price
in
\$1000's



size in feet²

you can rotate this figure

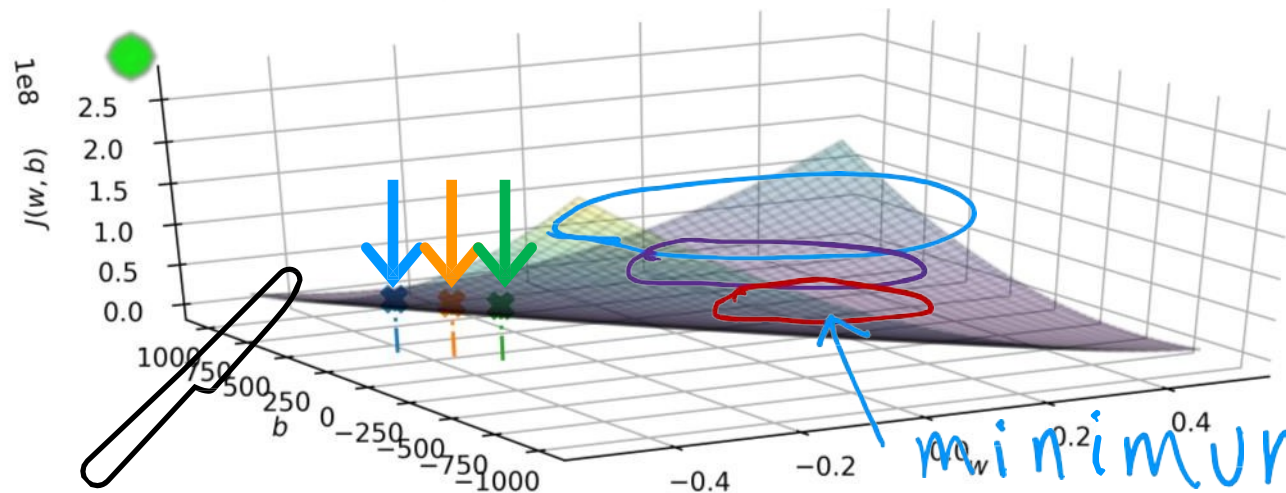
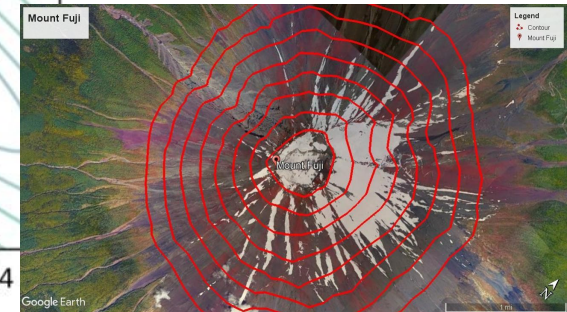
click within this plot to add points



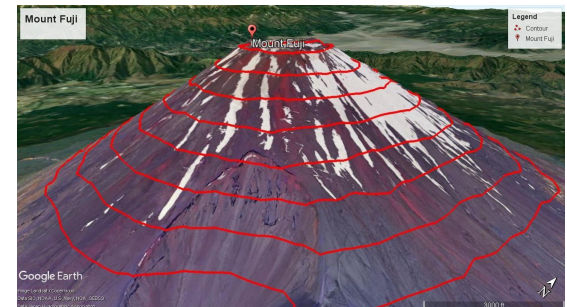
w

minimum

same J



minimum

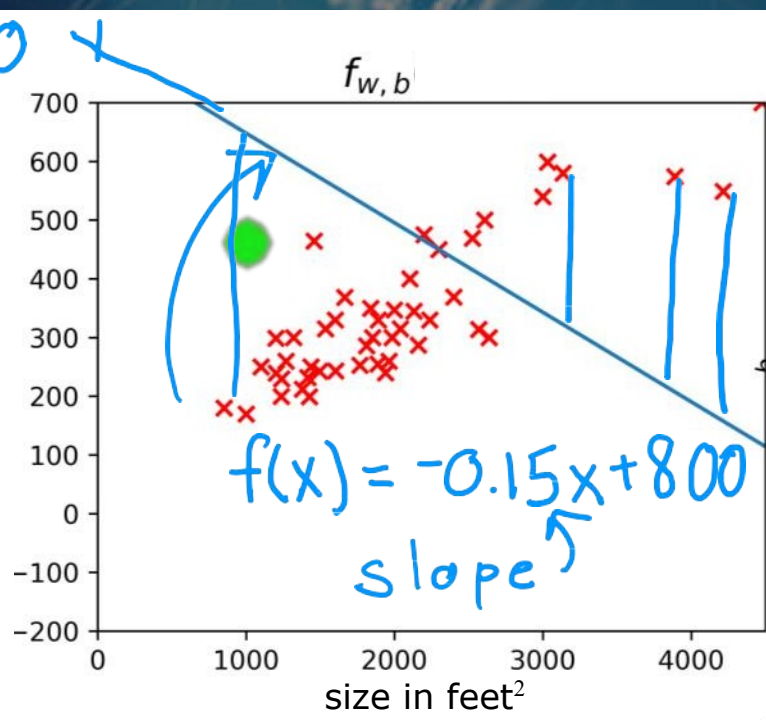




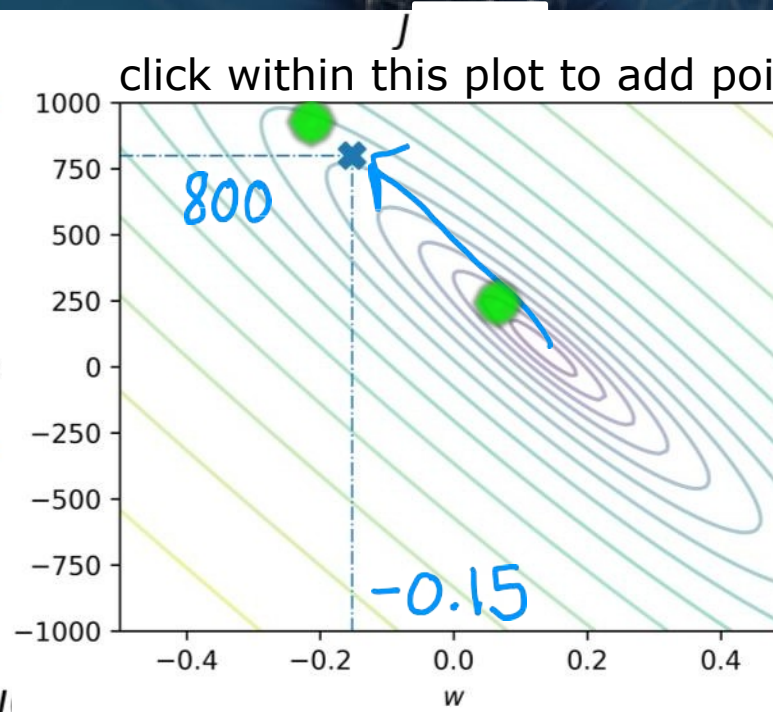
Visualization examples

800

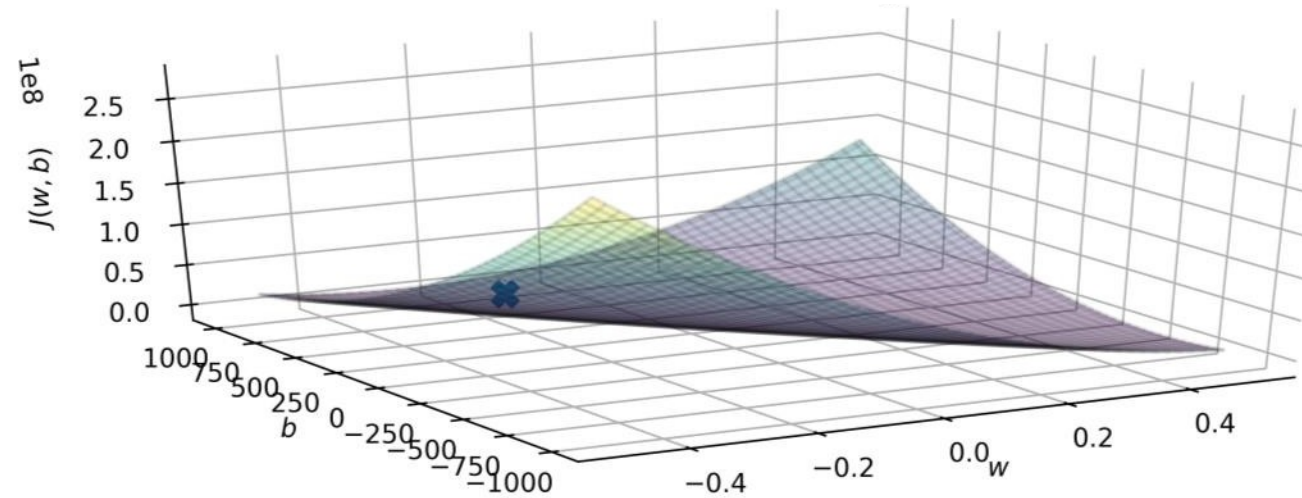
price
in
\$1000's



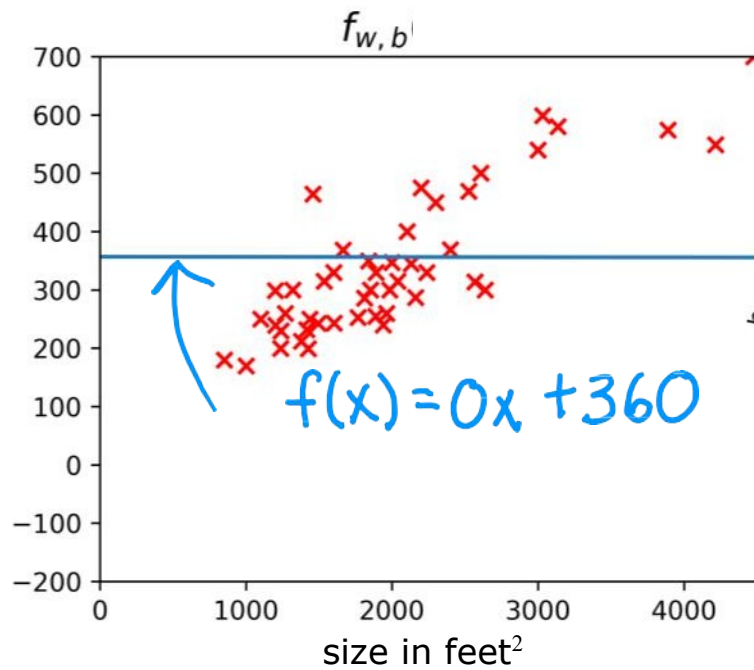
click within this plot to add points



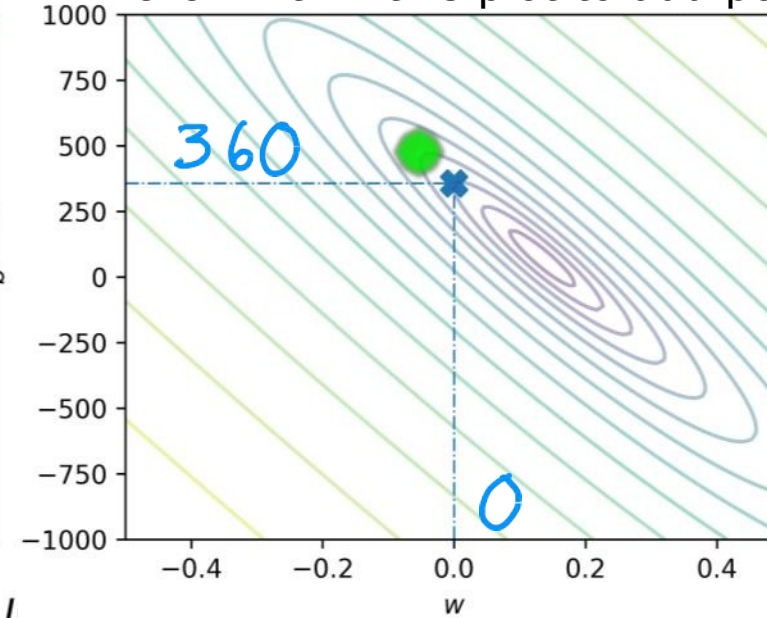
you can rotate this figure



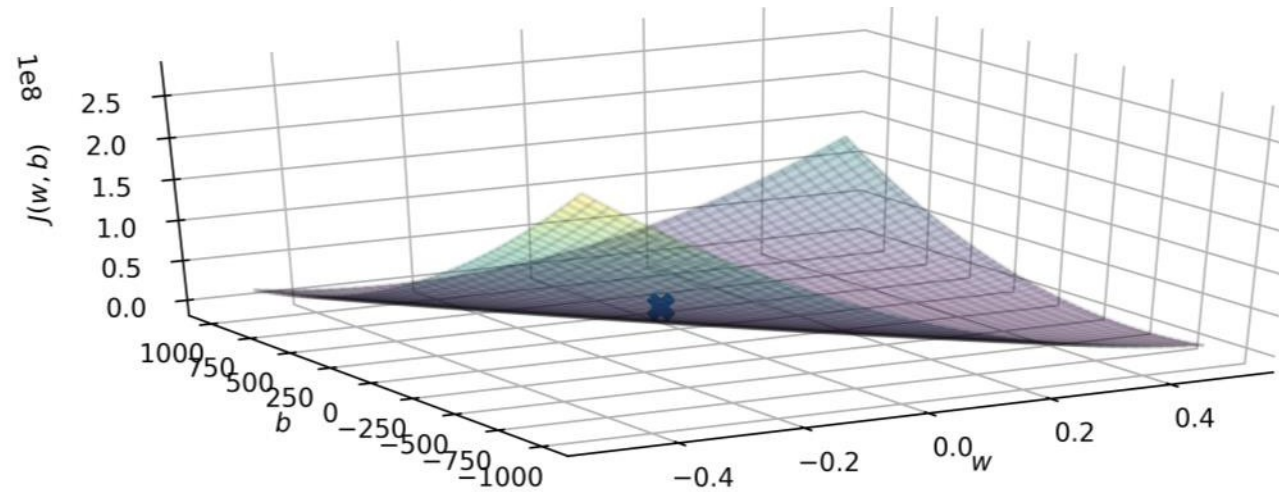
price
in
\$1000's



click within this plot to add points

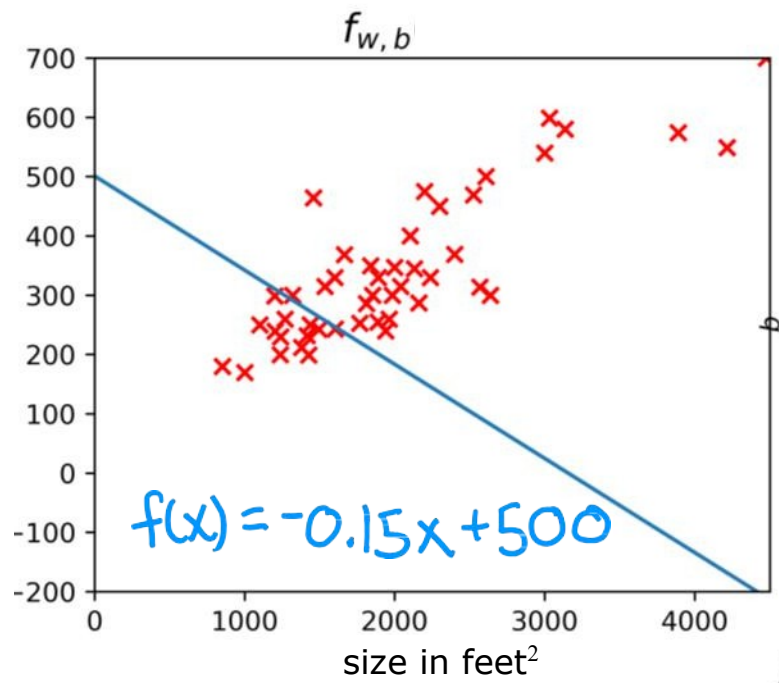


you can rotate this figure

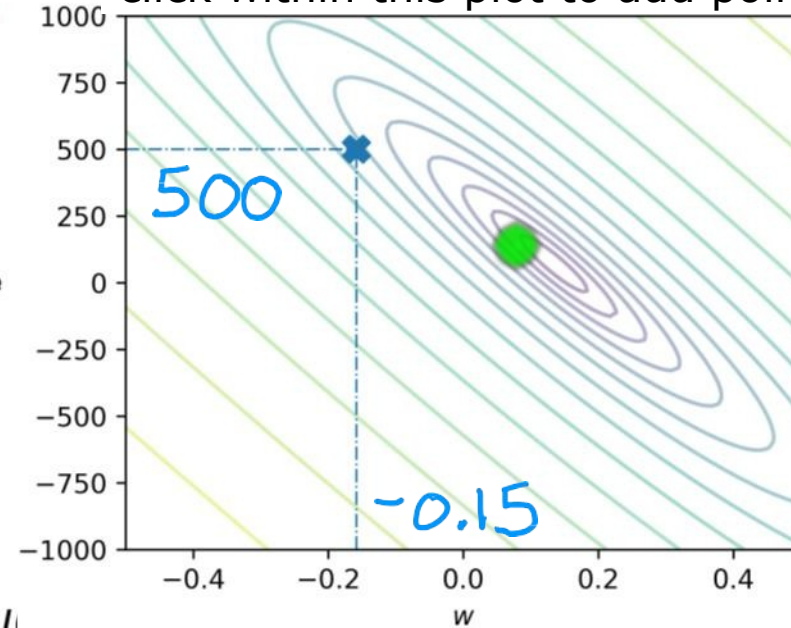




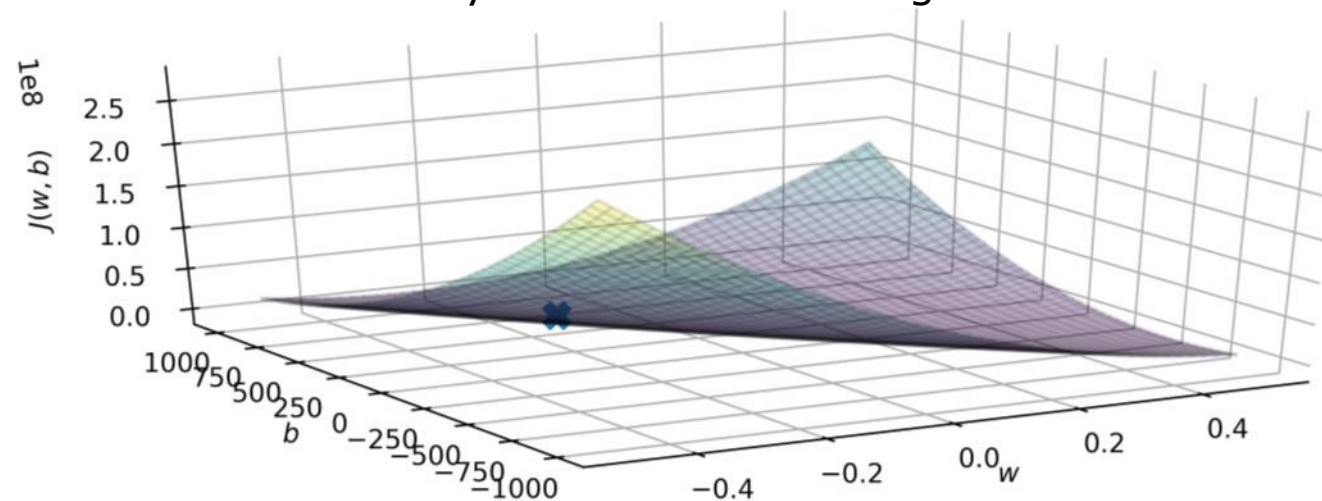
price
in
\$1000's



click within this plot to add points

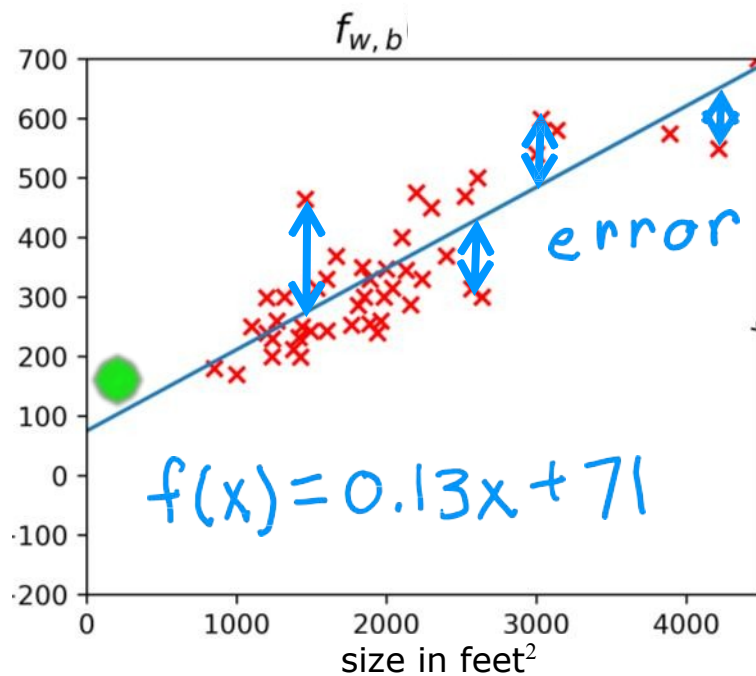


you can rotate this figure

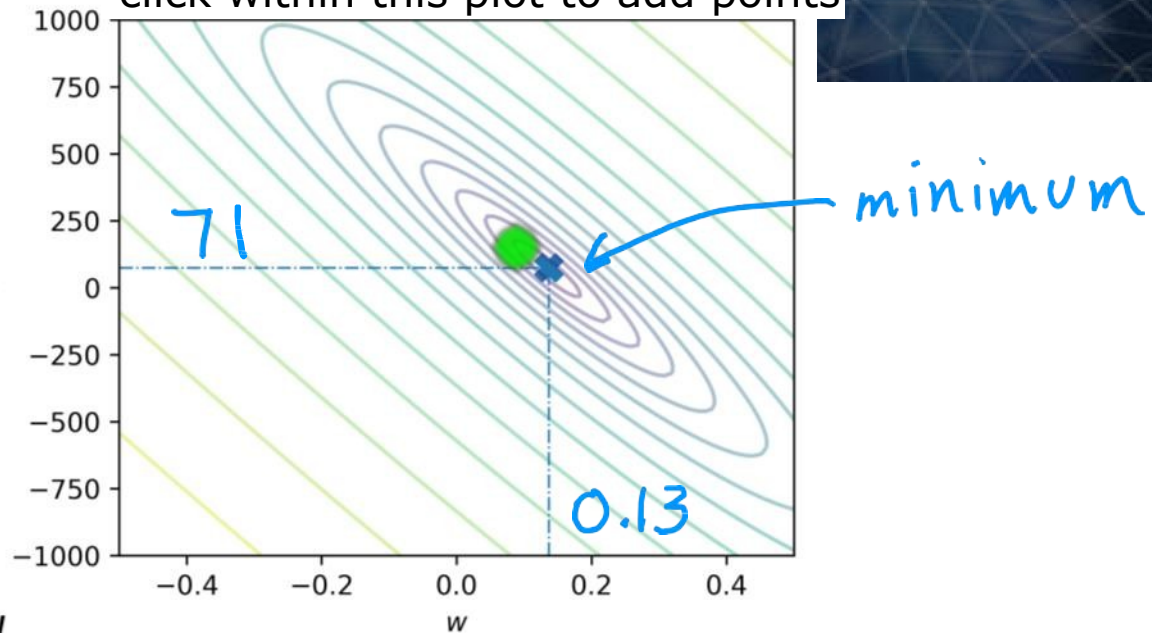




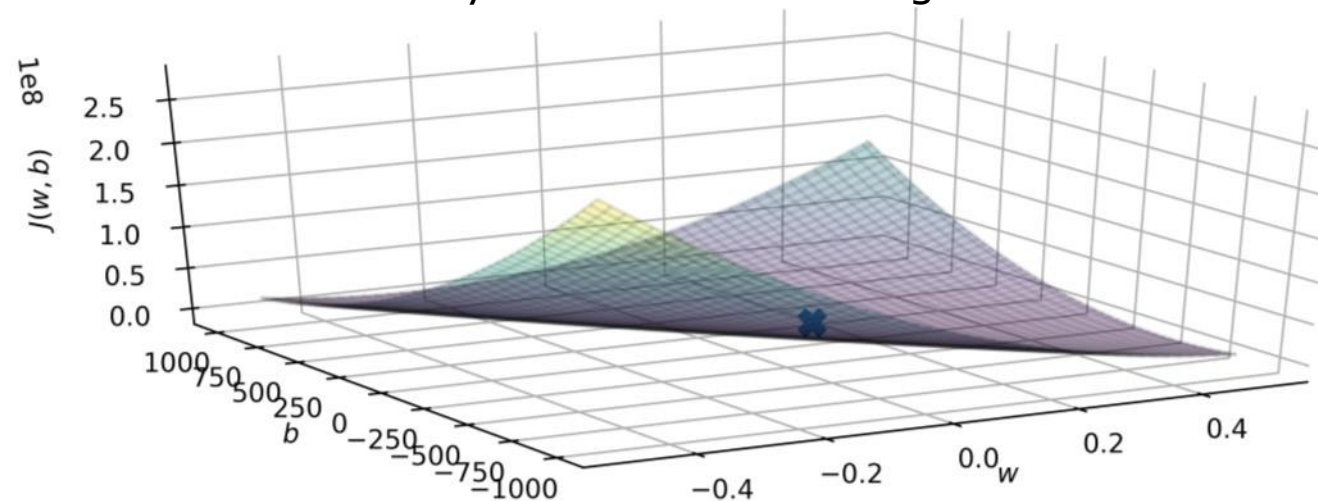
price
in
\$1000's



click within this plot to add points



you can rotate this figure





Q & A



Intuition Behind Derivatives

Derivatives

$$f(x) = 4x$$

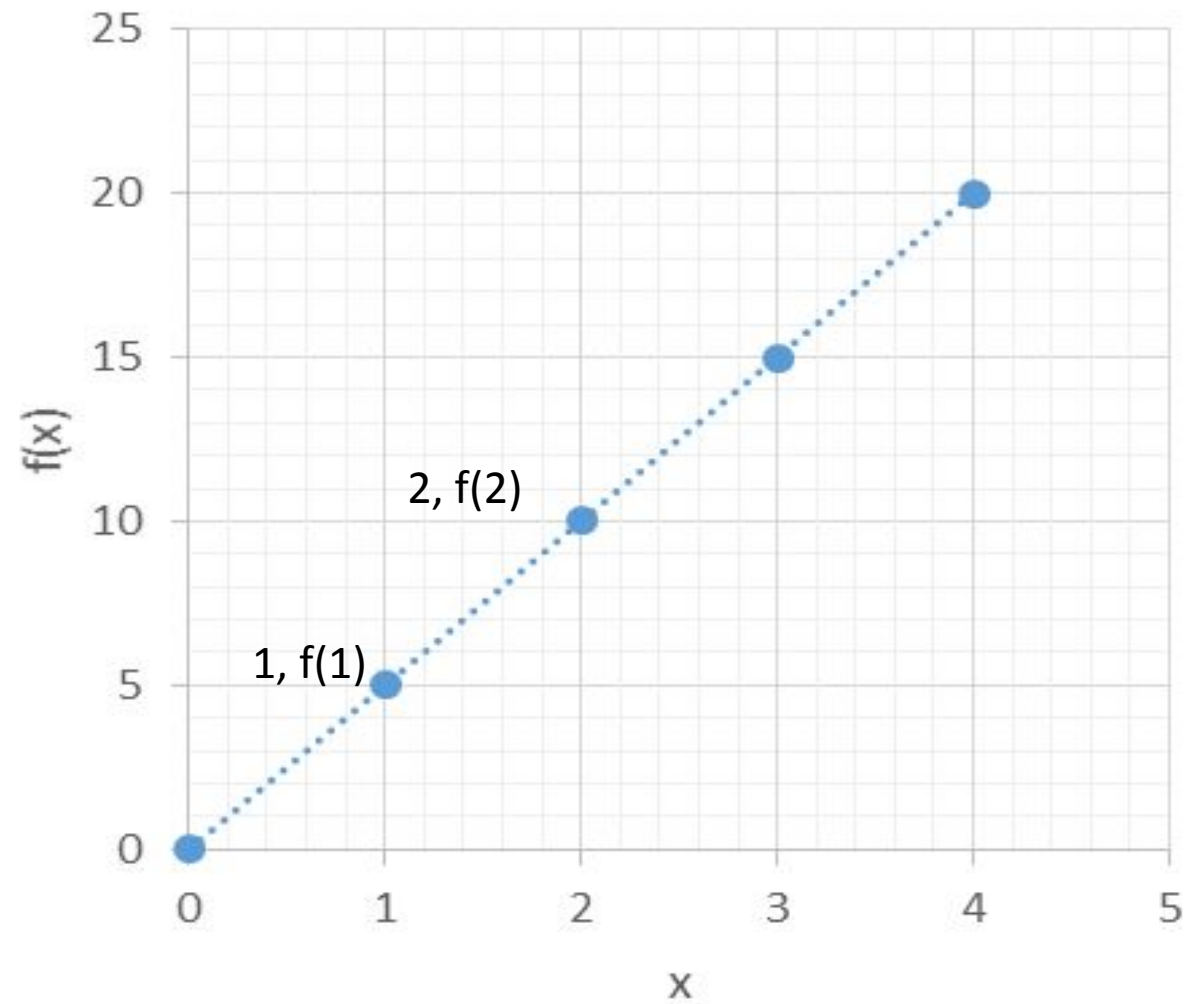
$$f(x) = x^3$$

$$f(x) = (x + 2)^4$$

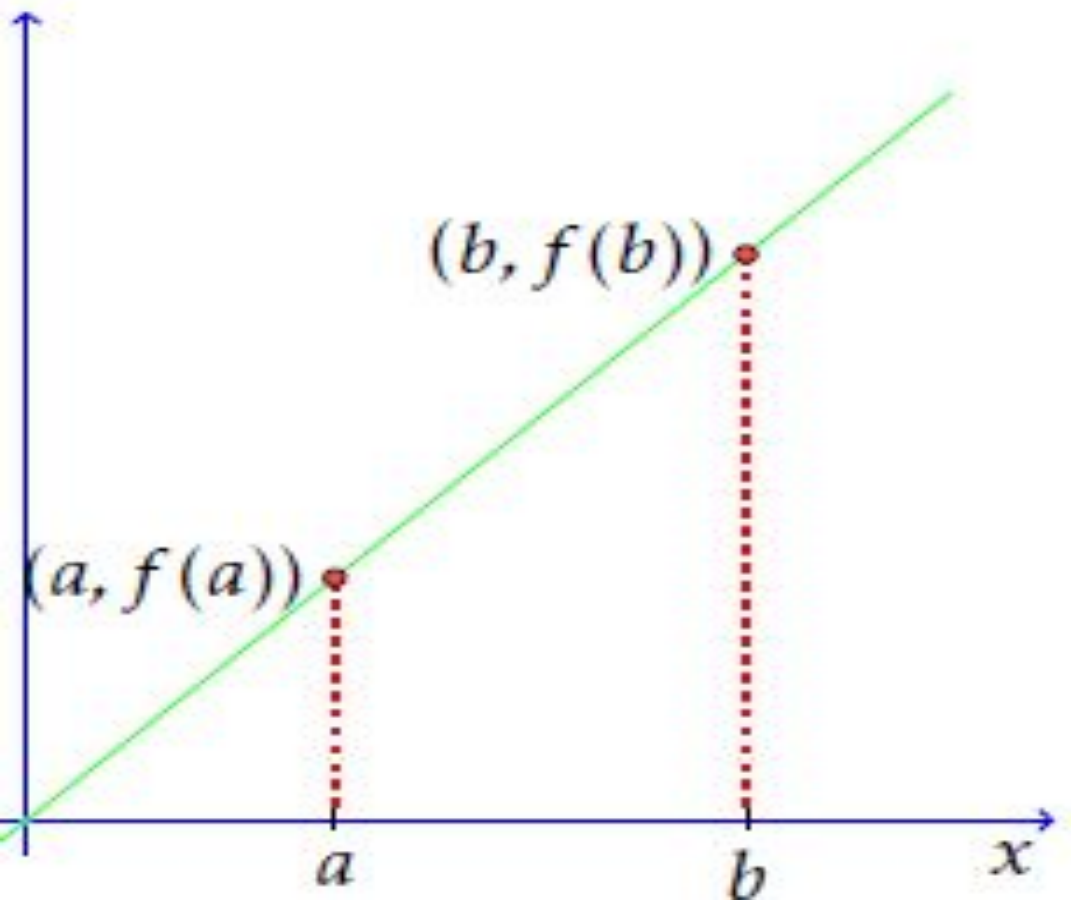
$$f(x) = (3x + 2y + 2)^2$$

Slope of line

x	$f(x)$
0	0
1	5
2	10
3	15
4	20

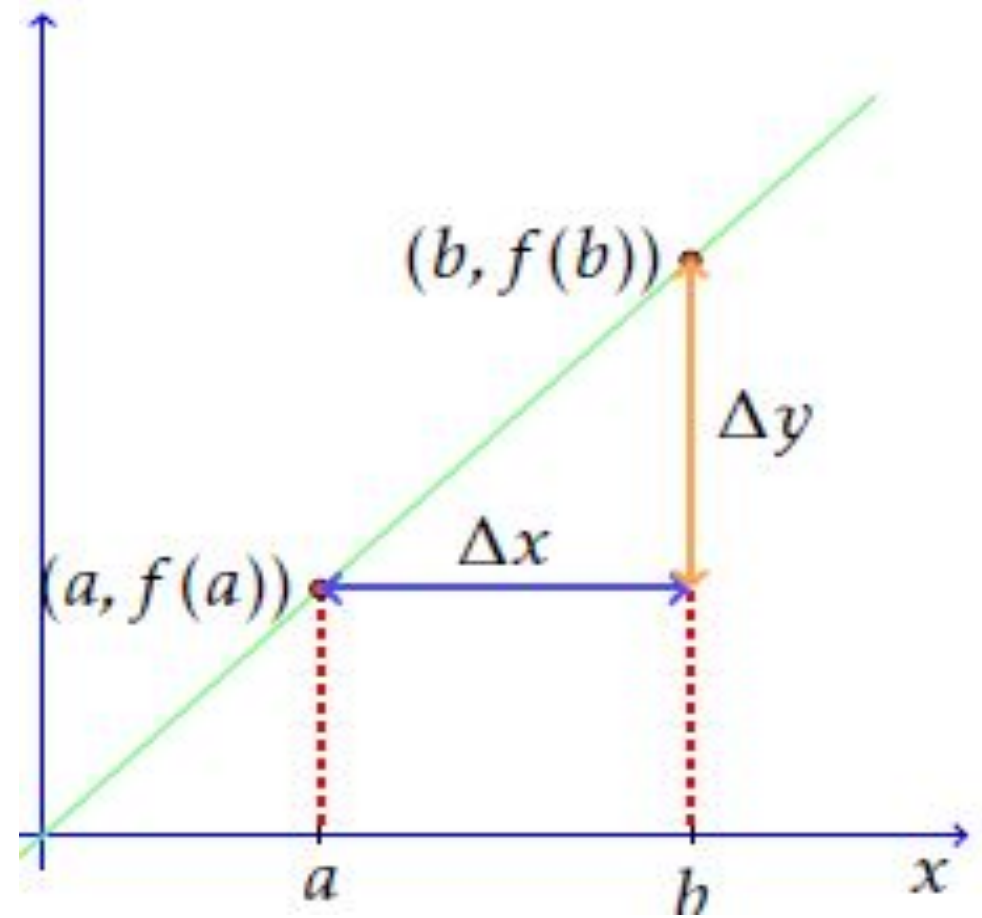


Slope of line



$$\Delta x = b - a$$

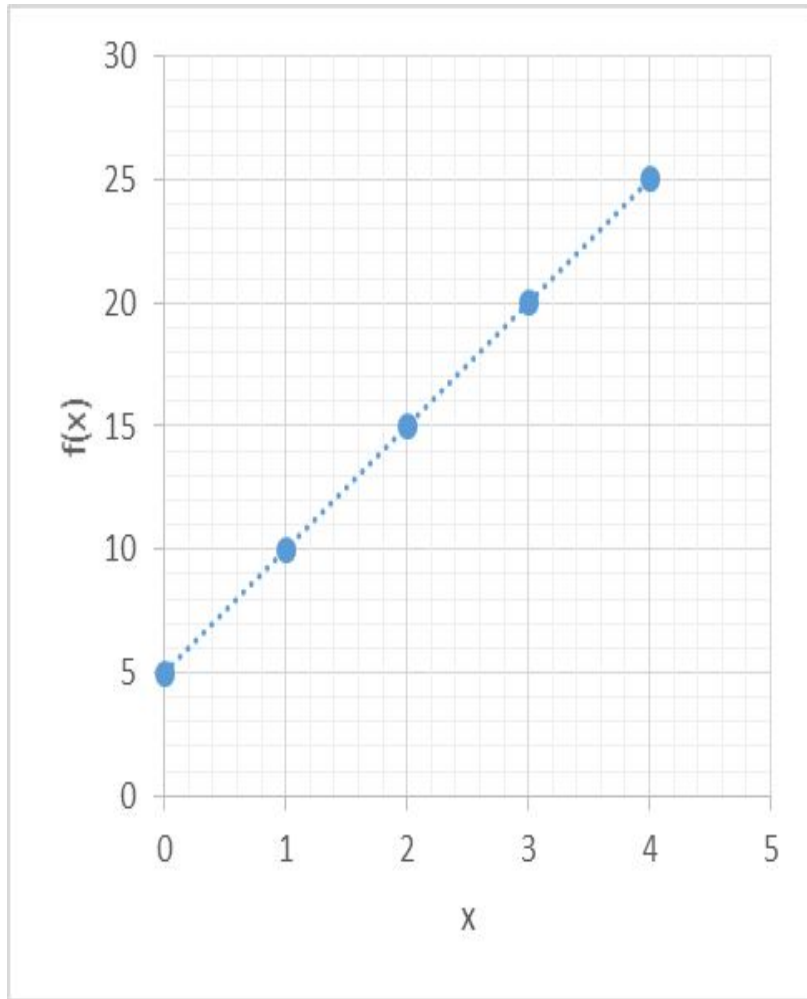
$$\Delta y = f(b) - f(a)$$



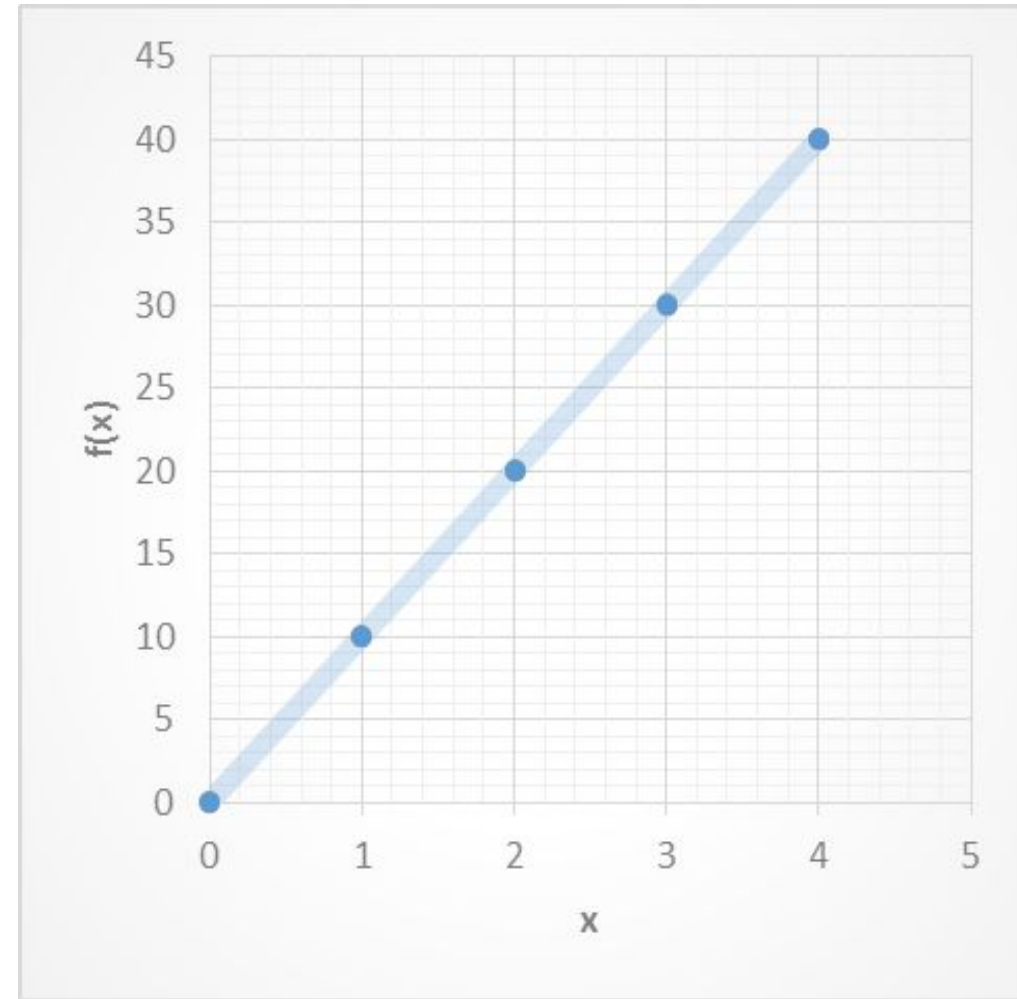
$$\text{slope} = \Delta y / \Delta x$$

Slope of line

x	$f(x)$
0	5
1	10
2	15
3	20
4	25



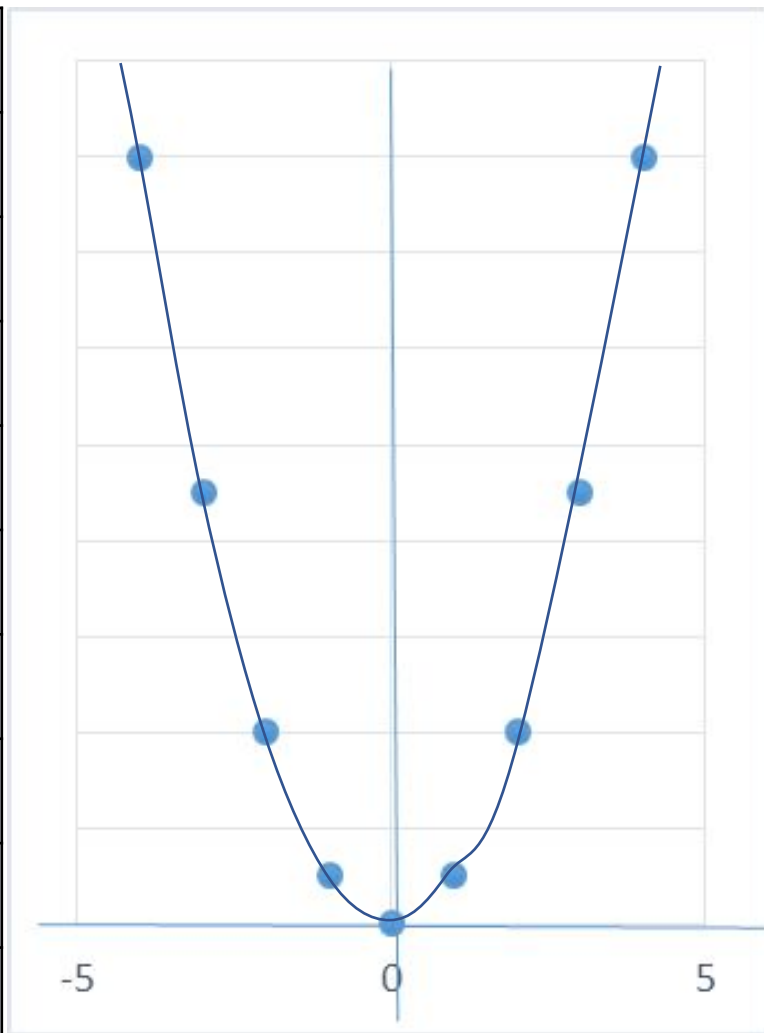
x	$f(x)$
0	0
1	10
2	20
3	30
4	40



The Slope of a Curve

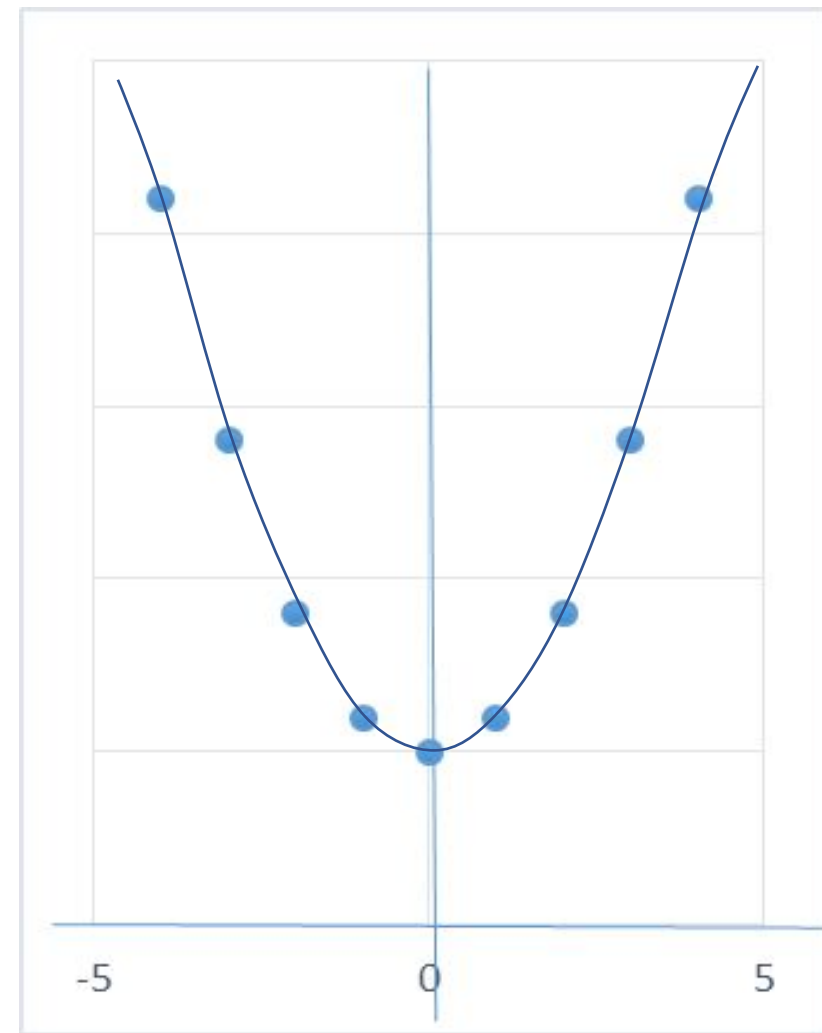
$$f(x) = x^2$$

x	f(x)
0	0
1	1
-1	1
2	4
-2	4
3	9
-3	9
4	16
-4	16



$$f(x) = x^2 + 5$$

x	f(x)
0	5
1	6
-1	6
2	9
-2	9
3	14
-3	14
4	21
-4	21





Q & A

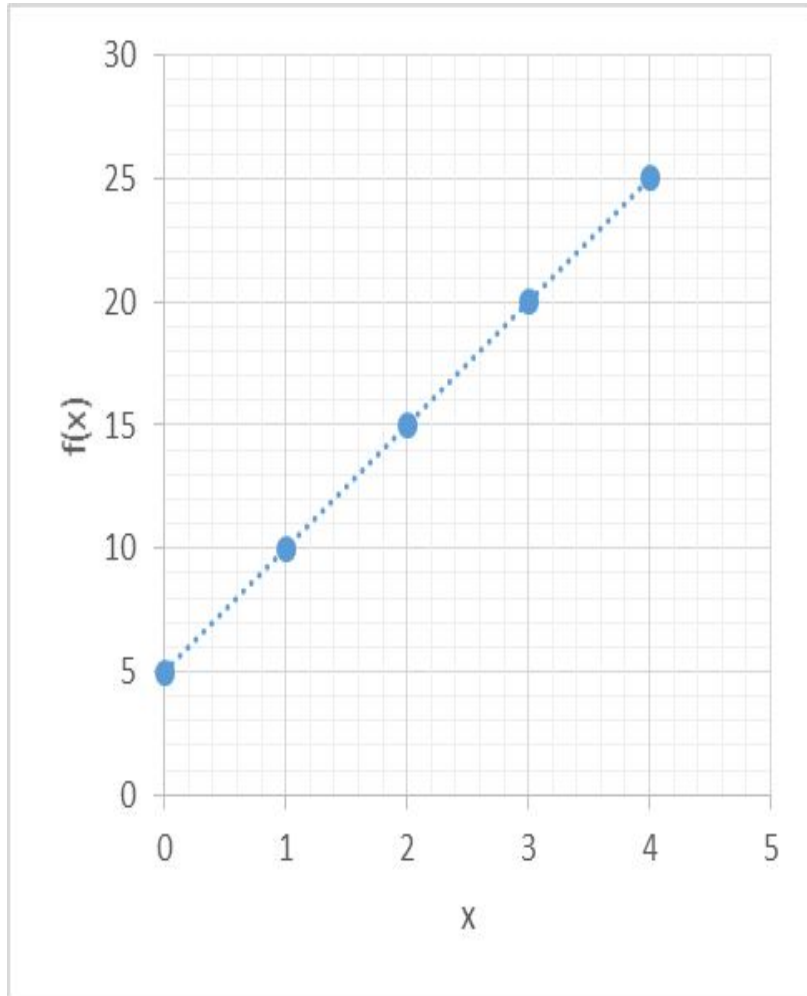


Finding Minima and Maxima

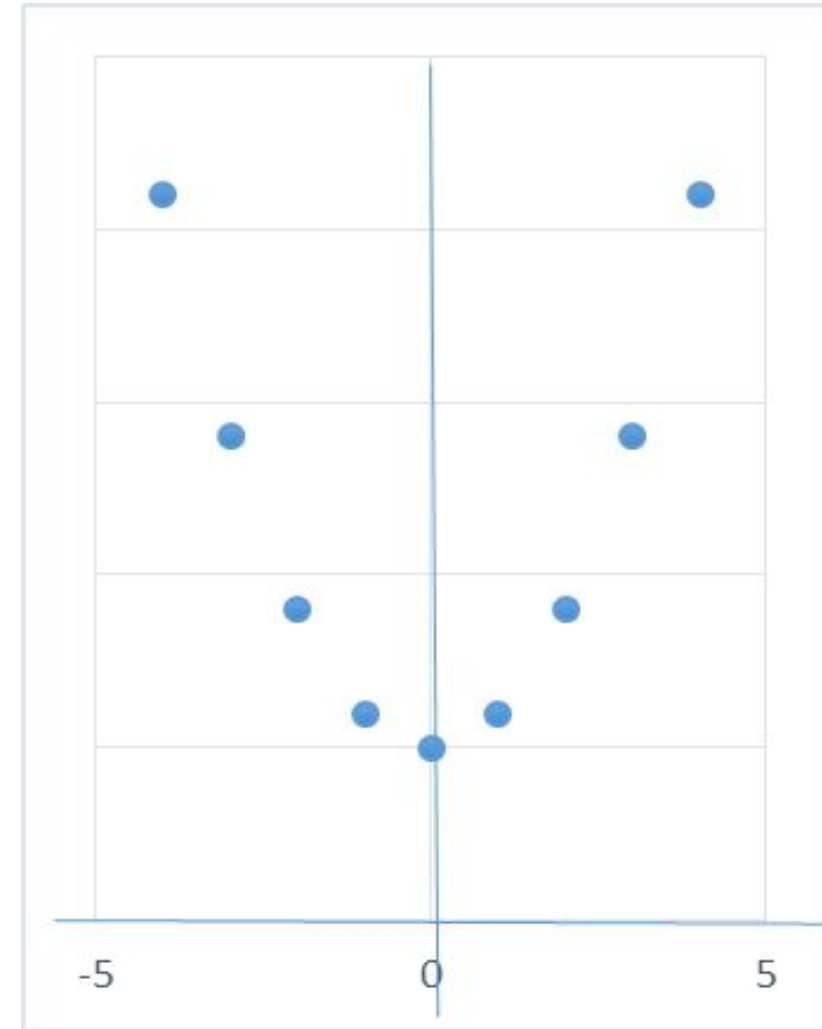
Finding Minima and Maxima

$$f(x) = x^2 + 5$$

x	f(x)
0	5
1	10
2	15
3	20
4	25



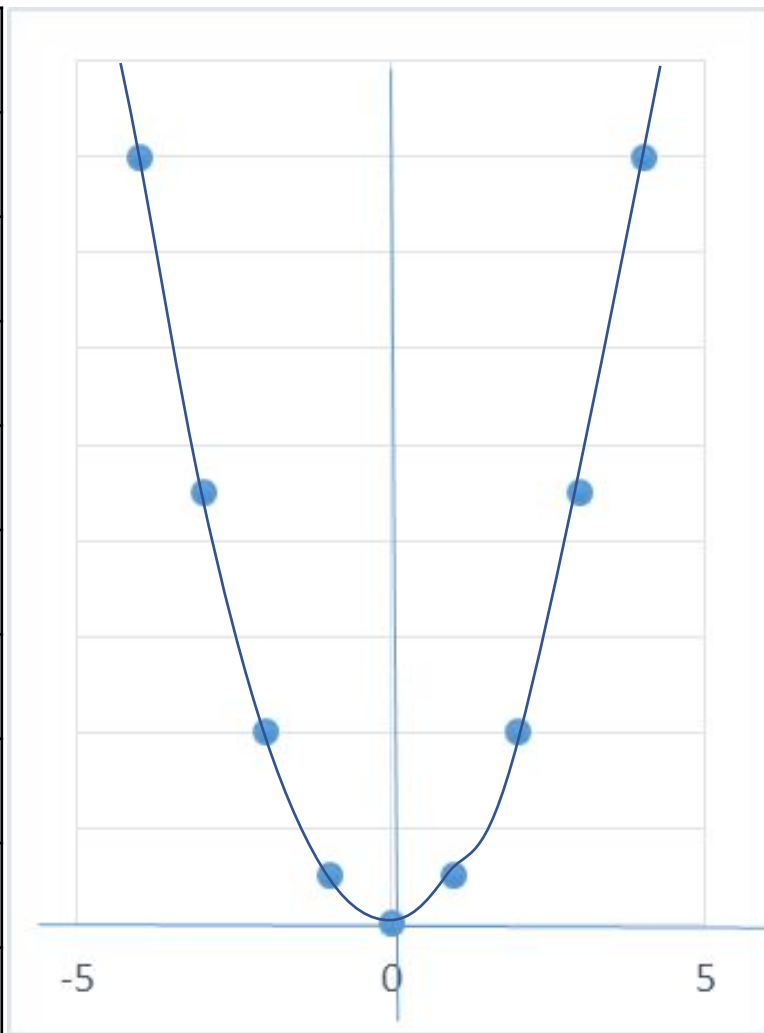
x	f(x)
0	5
1	6
-1	6
2	9
-2	9
3	14
-3	14
4	21
-4	21



Finding Minima and Maxima

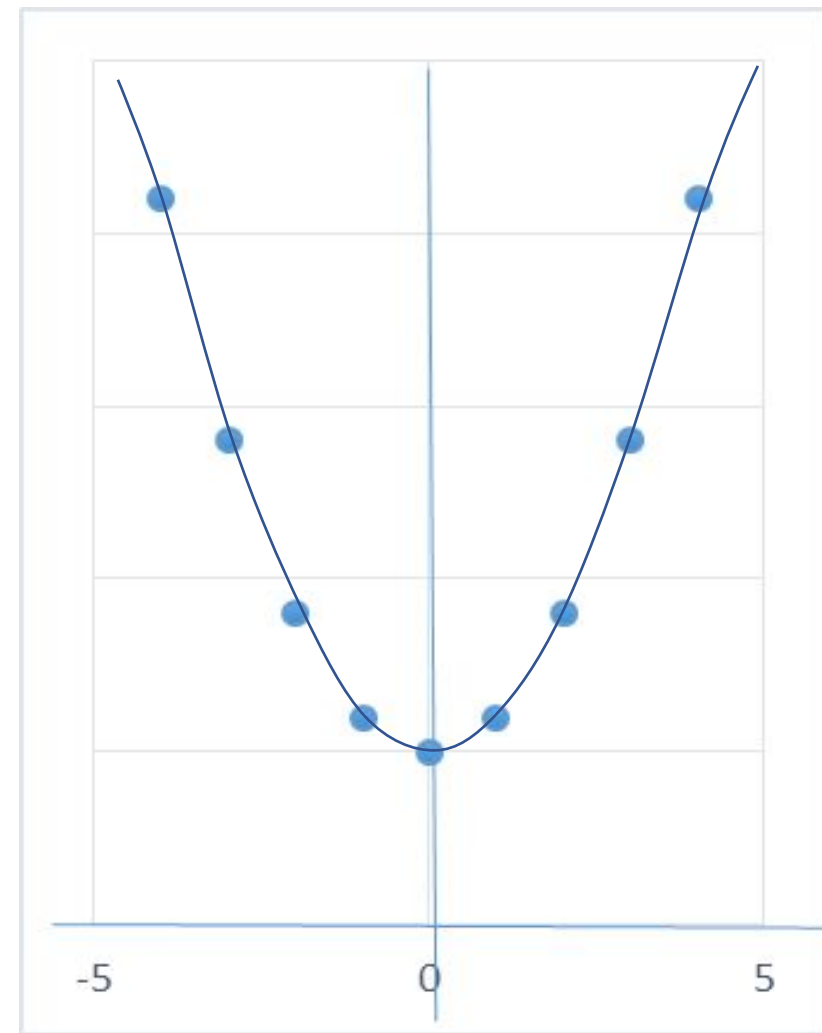
$$f(x) = x^2$$

x	f(x)
0	0
1	1
-1	1
2	4
-2	4
3	9
-3	9
4	16
-4	16

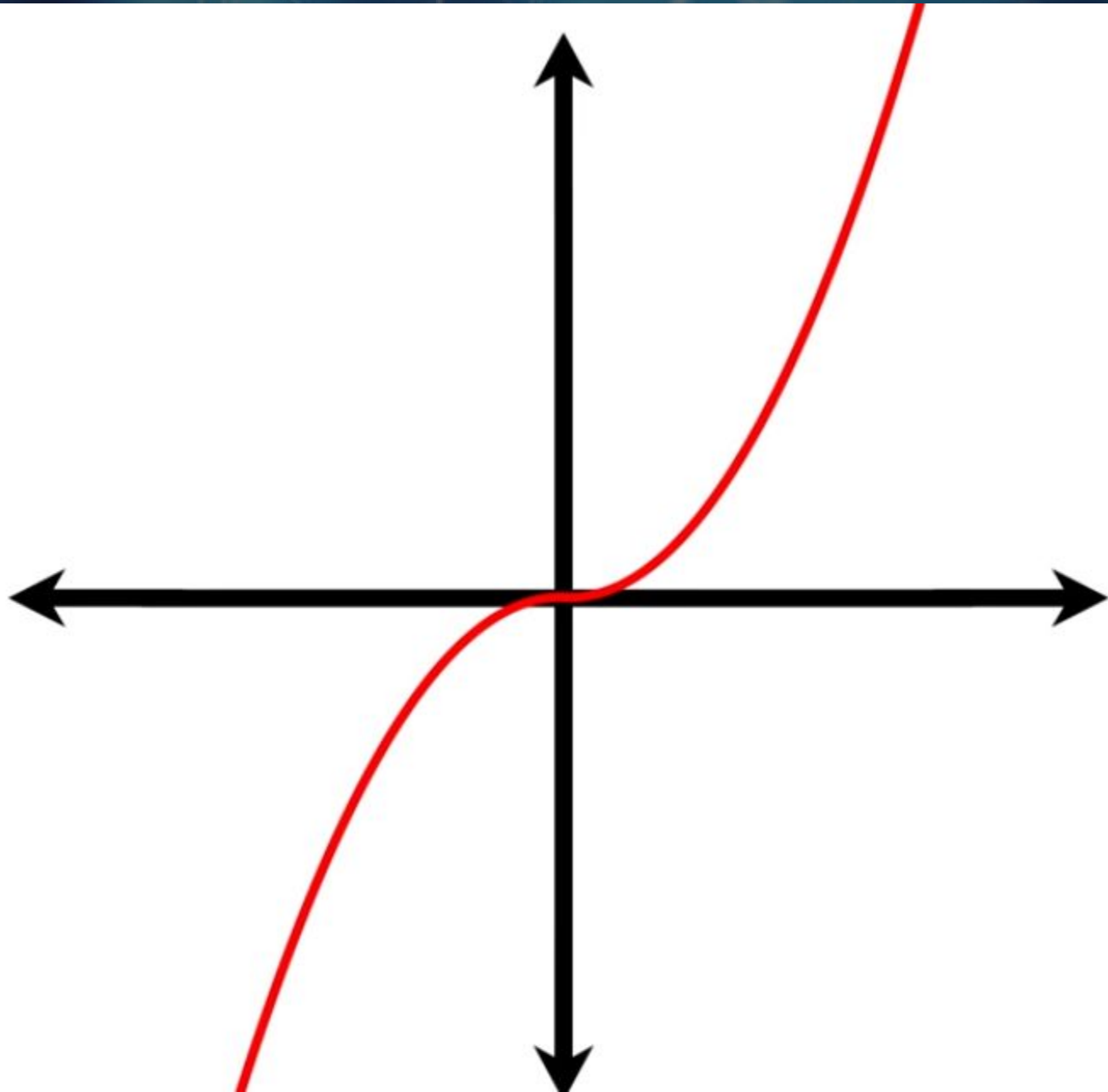


$$f(x) = x^2 + 5$$

x	f(x)
0	5
1	6
-1	6
2	9
-2	9
3	14
-3	14
4	21
-4	21

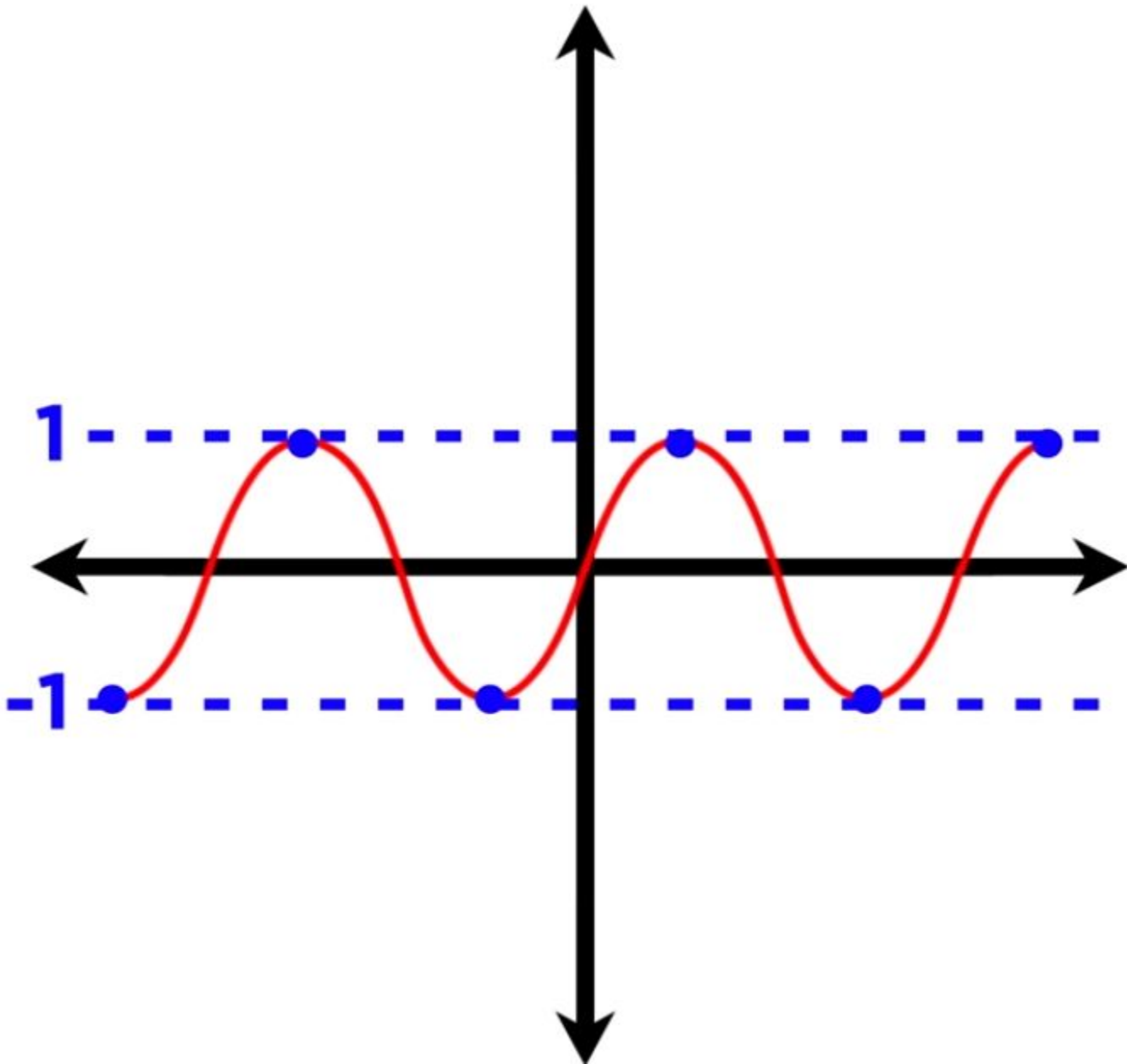


Finding Minima and Maxima



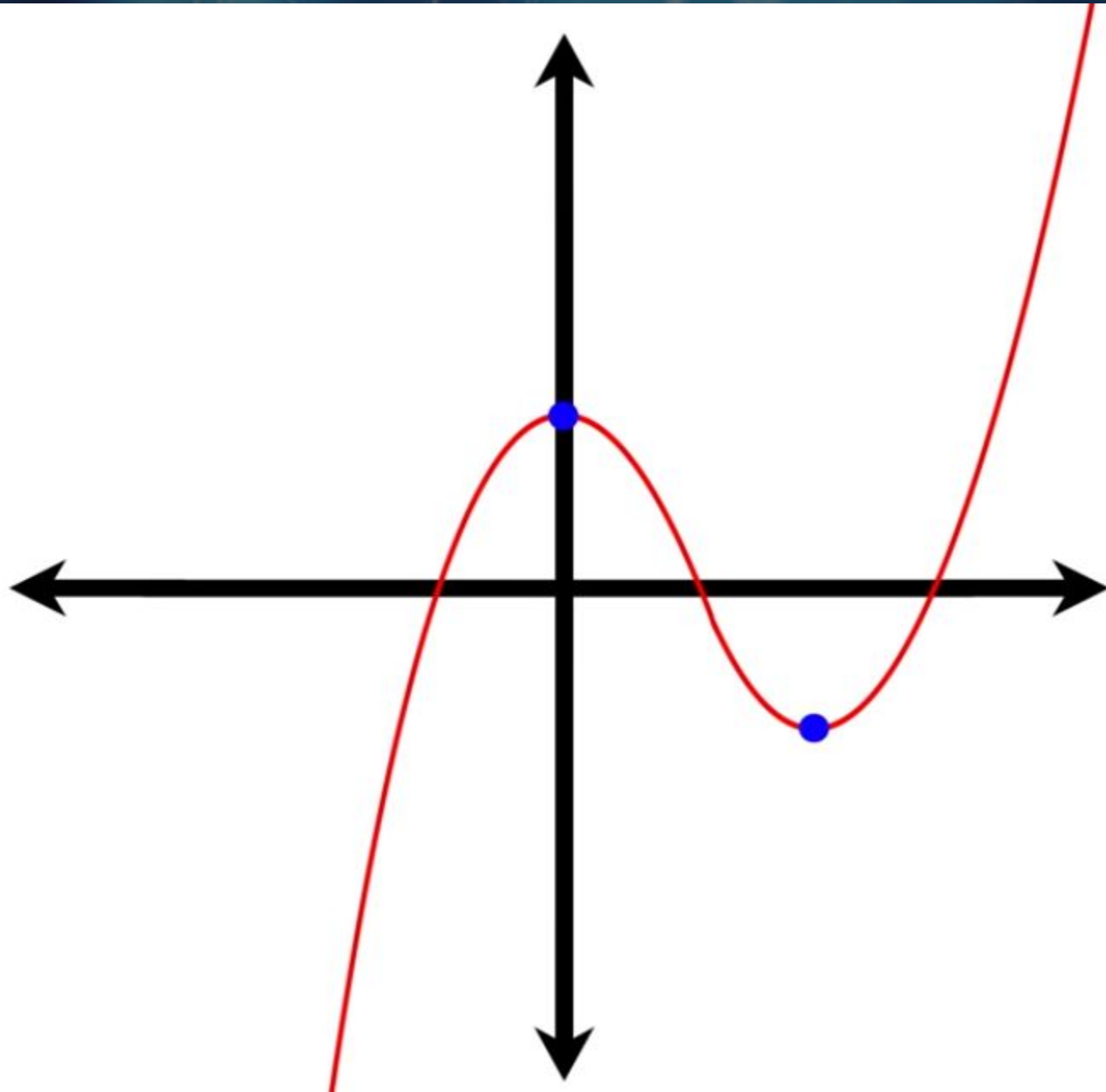
$f(x) = x^3$ has no
maxima/minima

Finding Minima and Maxima



$f(x) = \sin(x)$ has an
absolute maximum
and absolute minimum

Finding Minima and Maxima



$$f(x) = x^3 - 3x^2 + 1$$

$$f'(x) = 3x^2 - 6x$$

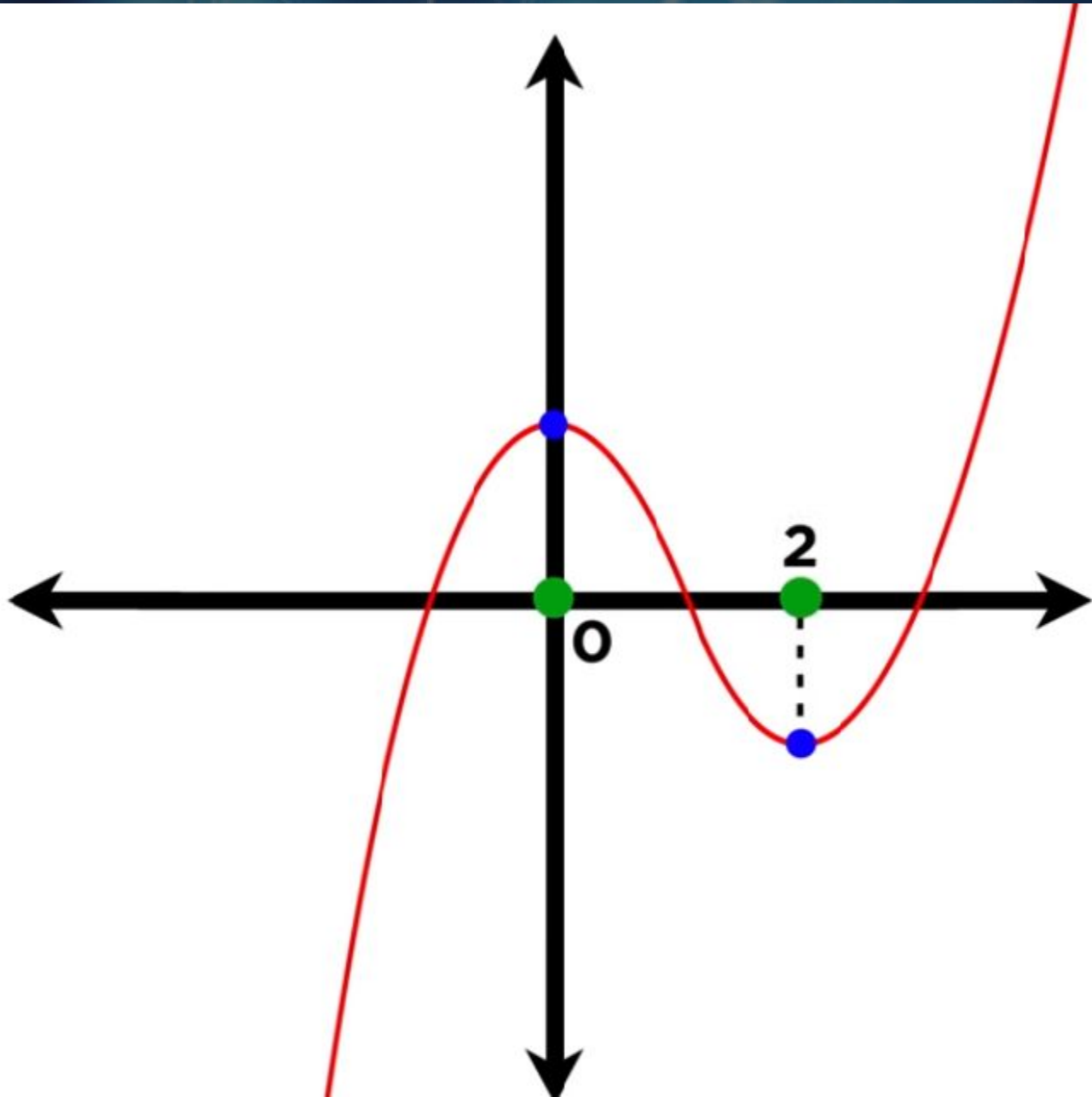
$$f'(x) = 3x(x-2)$$

$$3x(x-2) = 0$$

$$x = 0, 2$$

$$f'(0) = 0, f'(2) = 0$$

Finding Minima and Maxima



$$f(x) = x^3 - 3x^2 + 1$$

$$f'(x) = 3x^2 - 6x$$

$$f'(x) = 3x(x-2)$$

$$3x(x-2) = 0$$

$$x = 0, 2$$

$$f'(0) = 0, f'(2) = 0$$



Q & A